# Finanças II – Corporativas (MPE Insper)

Exercise list R3 — A frictionless benchmark, taxes, distress and static trade-off

*Worked solutions*

Statements are in `Listas/lista-r3-mm-taxes-tradeoff.md` and each is reproduced right before its solution; each section cites the exercise by number and printed page. Editions cited are the local copies: Berk, DeMarzo & Harford, *Fundamentals of Corporate Finance*, 2nd ed. (2011); Ross, Westerfield, Jordan & Lamb, *Fundamentos de Administração Financeira*, 9th ed. (2013); Koski, Higgins & Mitton, *Analysis for Financial Management*, 13th ed. (2023); Grinblatt & Titman, *Financial Markets and Corporate Strategy*, 2nd ed. (2002); Amaro de Matos, *Theoretical Foundations of Corporate Finance* (2001). Every number below is reproduced by `lista-r3-mm-taxes-tradeoff-solutions_check.py`. Amounts are in dollars unless a section says "millions". Each section follows the same order: what is asked, approach, notation, steps (with the answer to each part boxed), intuition, common mistakes, book answer and reference. Sections 16–18 solve *original* exercises on the bank capital debate (no course book has one); they have no book answer and cite Prof. Ricca's slides instead of a book page.

## 1. Two-outcome project financed with equity or safe debt — Berk 16.1, p. 490

**Statement.**

**Source:** Berk, Ch. 16, Exercise 16.1, p. 490 (PDF 521) · Answer in book: no

Consider a project with free cash flows in one year of $130,000 or $180,000, with each outcome being equally likely. The initial investment required for the project is $100,000, and the project's cost of capital is 20%. The risk-free interest rate is 10%.

a. What is the NPV of this project?

b. Suppose that to raise the funds for the initial investment, the project is sold to investors as an all-equity firm. The equity holders will receive the cash flows of the project in one year. How much money can be raised in this way—that is, what is the initial market value of the unlevered equity?

c. Suppose the initial $100,000 is instead raised by borrowing at the risk-free interest rate. What are the cash flows of the levered equity, and what is its initial value according to MM?

**Context needed to solve it** *From Berk Ch. 16, Problems, section "Capital Structure in Perfect Capital Markets" (instruction that heads this problem section), p. 490 (PDF 521):*

For problems in this section assume no taxes or distress costs.

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**What is asked.** (a) The NPV of the project: what it is worth today minus what it costs. (b) How much cash the entrepreneur raises by selling the whole project to equity investors, i.e. the market value today of an all-equity claim on the project's cash flows. (c) If the $100,000 is borrowed at the risk-free rate instead, what the shareholders receive in each state next year, and what that levered equity is worth today according to MM.

**Approach.** Part (a) is a plain NPV: discount the expected cash flow at the project's cost of capital (20%), which prices its risk. Part (b) uses the law of one price: unlevered equity receives exactly the project's cash flows, so it is worth exactly the project's PV. Part (c) is *MM Proposition I* in a perfect market (the section heading rules out taxes and distress costs): debt plus levered equity are claims to the same cash flows, so $E_L + D = V_U$. A last step checks the answer with *MM Proposition II*, which says how much more the riskier levered equity must earn.

**Notation.**

- $C_{\text{low}} = 130{,}000$, $C_{\text{high}} = 180{,}000$: project free cash flow at $t = 1$ in the two states; $p = \frac{1}{2}$: probability of the low state.
- $I = 100{,}000$: initial investment; $r_U = 0.20$: project (unlevered) cost of capital; $r_f = 0.10$: risk-free rate.

- $V_U$: value of the project's cash flows (= unlevered equity $E_U$); $NPV = V_U - I$.
- $D = 100{,}000$: amount borrowed at $r_f$; $E_L$: value of levered equity; $r_E$: its expected return; $r_D = r_f$.

**Step 1 — Expected cash flow.** The cost of capital is applied to the expected cash flow, so first average the two equally likely outcomes:

$$\begin{aligned}\mathbb{E}[C] &= p\,C_{\text{low}} + (1-p)\,C_{\text{high}} \\ &= (0.5)(130{,}000) + (0.5)(180{,}000) \\ &= 65{,}000 + 90{,}000 \\ &= 155{,}000.\end{aligned} \tag{1.1}$$

**Step 2 — (a) Present value and NPV.** Discount (1.1) one year at the project's cost of capital and subtract the investment:

$$\begin{aligned}V_U &= \frac{\mathbb{E}[C]}{1+r_U} \\ &= \frac{(155{,}000)}{(1.20)} \\ &= 129{,}166.67,\end{aligned} \tag{1.2}$$

$$\begin{aligned}NPV &= V_U - I \\ &= (129{,}166.67) - (100{,}000) \\ &= 29{,}166.67.\end{aligned} \tag{1.3}$$

**Answer (a).** $NPV = \$29{,}166.67$.

**Step 3 — (b) Unlevered equity.** All-equity holders receive exactly the project's cash flow in each state, so by the law of one price their claim is worth the PV in (1.2):

$$E_U = V_U = 129{,}166.67.$$

Selling the equity raises \$129,166.67, which exceeds $I$ by the NPV: the entrepreneur keeps \$29,166.67 after paying for the investment.

**Answer (b).** The unlevered equity is worth (and raises) \$129,166.67.

**Step 4 — (c) Repayment and the levered equity's cash flows.** The loan is repaid with interest in one year:

$$\begin{aligned}D(1+r_f) &= (100{,}000)(1.10) \\ &= 110{,}000.\end{aligned} \tag{1.4}$$

Since $110{,}000 < C_{\text{low}} = 130{,}000$, the firm repays in both states: the debt is indeed riskless, which justifies borrowing at $r_f$. Equity receives what is left:

$$\begin{aligned}\text{low state:} \quad & C_{\text{low}} - 110{,}000 = 130{,}000 - 110{,}000 = 20{,}000, \\ \text{high state:} \quad & C_{\text{high}} - 110{,}000 = 180{,}000 - 110{,}000 = 70{,}000.\end{aligned}$$

**Step 5 — (c) Value by MM I.** With no taxes or distress costs, MM I says the debt and the levered equity together are worth the unlevered firm, $E_L + D = V_U$. Solving for $E_L$:

$$\begin{aligned}E_L &= V_U - D \\ &= (129{,}166.67) - (100{,}000) \\ &= 29{,}166.67.\end{aligned} \tag{1.5}$$

**Step 6 — Consistency with MM II.** The expected equity cash flow is $(0.5)(20{,}000) + (0.5)(70{,}000) = 10{,}000 + 35{,}000 = 45{,}000$, so the expected return implied by (1.5) is

$$\begin{aligned} r_E &= \frac{45{,}000}{E_L} - 1 \\ &= \frac{45{,}000}{(29{,}166.67)} - 1 \\ &= 0.542857. \end{aligned} \tag{1.6}$$

MM II without taxes, $r_E = r_U + \frac{D}{E_L}(r_U - r_D)$, needs the debt-equity ratio

$$\begin{aligned} \frac{D}{E_L} &= \frac{(100{,}000)}{(29{,}166.67)} = 3.428571, \\ r_E &= (0.20) + (3.428571)(0.20 - 0.10) \\ &= (0.20) + (3.428571)(0.10) \\ &= 0.542857, \end{aligned}$$

the same as (1.6).

**Answer (c).** Levered equity pays \$20,000 (low) or \$70,000 (high); by MM it is worth \$29,166.67 (expected return 54.29%).

**Intuition.** Borrowing slices the same cash flows differently; it does not change their total value. The entrepreneur captures the same NPV, \$29,166.67, either as the excess of the equity sale over $I$ or as the value of the levered equity left after the loan. The levered equity bears all the project's risk on a smaller base, so it must offer a higher expected return (54.29% against 20%).

**Common mistakes.**

- Discounting the levered equity's expected cash flow at the project's 20%: $45{,}000/1.20 = 37{,}500$ overvalues it, because levered equity is riskier than the project.
- Subtracting only the principal (\$100,000) from the cash flows and forgetting the \$10,000 of interest.
- Not checking that the debt is riskless: the risk-free rate is the right borrowing rate only because \$110,000 is below the worst cash flow.

**Book answer.** None in the book.

**Ref:** Berk Ch. 16, §16.2 "Capital Structure in Perfect Capital Markets", pp. 464–472 (MM I, MM II).

## 2. One-time dividend or repurchase in a perfect market — Berk 17.7, p. 528

**Statement.**

**Source:** Berk, Ch. 17, Exercise 17.7, p. 528 (PDF 559) · Answer in book: no

Natsam Corporation has \$250 million of excess cash. The firm has no debt and 500 million shares outstanding with a current market price of \$15 per share. Natsam's board has decided to pay out this cash as a one-time dividend.

a. What is the ex-dividend price of a share in a perfect capital market?

b. If the board instead decided to use the cash to do a one-time share repurchase, in a perfect capital market, what is the price of the shares once the repurchase is complete?

c. In a perfect capital market, which policy in part (a) or (b) makes investors in the firm better off?

**What is asked.** (a) The share price right after the \$250 million is paid out as a dividend. (b) The share price after the same cash is used to buy back shares instead. (c) Which of the two policies leaves investors richer in a perfect capital market.

**Approach.** In a perfect market the share price equals the firm's market value per share, and that value includes the excess cash. Paying the cash out shrinks the firm by exactly the cash paid. With a dividend the number of shares is fixed, so the price falls; with a repurchase at a fair price the number of shares falls in proportion, so the price is unchanged. This is *MM payout irrelevance*: compare total wealth per share (stock plus cash), not prices.

**Notation.**

- $N = 500$ million shares; $P_0 = 15$: cum-payout share price; $C = 250$ million: excess cash paid out.
- $d$: dividend per share; $P_{\text{ex}}$: ex-dividend price; $n_R$: shares repurchased (millions); $P_R$: post-repurchase price.

**Step 1 — Market value of the firm.** With no debt, the firm's value is its equity, cash included:

$$NP_0 = (500)(15) = 7{,}500 \text{ million},$$
$$\text{of which operating assets} = 7{,}500 - 250 = 7{,}250 \text{ million}.$$

**Step 2 — (a) Ex-dividend price.** The dividend per share is the cash divided by the shares; after it is paid, the firm holds only the operating assets:

$$d = \frac{C}{N} = \frac{(250)}{(500)} = 0.50, \tag{2.1}$$

$$\begin{aligned} P_{\text{ex}} &= \frac{NP_0 - C}{N} \qquad (2.2)\\ &= \frac{(7{,}500) - (250)}{(500)}\\ &= \frac{7{,}250}{500}\\ &= 14.50. \end{aligned}$$

The price falls by exactly the dividend: $15 - 0.50 = 14.50$.

**Answer (a).** $P_{\text{ex}} = \$14.50$.

**Step 3 — (b) Repurchase.** In a perfect market the shares are bought at the fair cum-payout price $P_0$, so the cash buys $C/P_0$ shares and the operating assets are spread over the remaining shares:

$$n_R = \frac{C}{P_0} = \frac{(250)}{(15)} = 16.6667, \tag{2.3}$$

$$\begin{aligned} P_R &= \frac{NP_0 - C}{N - n_R} \qquad (2.4)\\ &= \frac{(7{,}500) - (250)}{(500) - (16.6667)}\\ &= \frac{7{,}250}{483.3333}\\ &= 15.00. \end{aligned}$$

**Answer (b).** $P_R = \$15.00$ (16.67 million shares repurchased).

**Step 4 — (c) Comparison of wealth.** Per original share, the dividend leaves stock plus cash of

$$P_{\text{ex}} + d = (14.50) + (0.50) = 15.00,$$

by (2.1)–(2.2); the repurchase leaves either a share worth 15.00 by (2.4) or 15.00 in cash for a tendered share. Wealth is \$15 per share under both.

**Step 5 — Homemade dividend.** An investor with 1,000 shares who prefers the dividend but gets the repurchase sells $500/15 = 33.33$ shares at \$15 to raise the \$500 the dividend would have paid, and keeps $1{,}000 - 33.33 = 966.67$ shares worth $(1{,}000 - 500/15)(15) = 15{,}000 - 500 = 14{,}500$: exactly the dividend outcome, \$14,500 in stock plus \$500 in cash.

**Answer (c).** Neither: wealth is \$15 per share either way, and each investor can convert one policy into the other at no cost.

**Intuition.** Paying out cash shrinks the firm by exactly the cash paid. With a dividend the price falls by the dividend; with a repurchase the share count falls instead and the price stays put. Any investor can create the other pattern for herself (sell shares, or reinvest the dividend), so payout policy is irrelevant (MM).

**Common mistakes.**

- Thinking the repurchase raises the price because there are fewer shares, forgetting that the \$250 million of cash leaves the firm too.
- Leaving the ex-dividend price at \$15: the share no longer carries the right to the cash.
- Comparing prices instead of wealth: \$14.50 < \$15 does not mean the dividend makes investors poorer, since they also hold \$0.50 of cash.

**Book answer.** None in the book.

**Ref:** Berk Ch. 17, §17.2 "Dividends Versus Share Repurchases in a Perfect Capital Market", pp. 503–508.

## 3. MM I with corporate taxes — Ross 16.14, p. 578

**Statement.**

**Source:** Ross, Cap. 16, Exercise 16.14, p. 578 (PDF 602) · Answer in book: no (selected answers in the web-only Apêndice C)

A Frederico & Co. espera um Lajir de \$92.000 por ano para sempre. A empresa pode tomar empréstimos a 9%. Atualmente, a Frederico não tem dívida, e o custo do capital próprio é 15%. Se a alíquota tributária for 35%, qual será o valor da empresa? Qual será o valor se a empresa tomar empréstimos de \$60.000 e usar a receita para recomprar ações?

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**What is asked.** The value of Frederico & Co. as it is (all equity), and its value if it borrows \$60,000 permanently and uses the money to buy back shares.

**Approach.** This is *MM Proposition I with corporate taxes*. The all-equity value is the after-tax perpetual cash flow capitalised at the unlevered cost of capital $R_U$. Perpetual debt creates a perpetual *interest tax shield* $T_C R_D D$; it is as risky as the debt, so it is discounted at $R_D$ and is worth $T_C D$. The levered value is $V_L = V_U + T_C D$; the buyback only changes who holds the claims.

**Notation.**

- $EBIT = 92{,}000$ per year forever; $R_U = 0.15$: unlevered cost of equity; $R_D = 0.09$: borrowing rate.
- $T_C = 0.35$: corporate tax rate; $D = 60{,}000$: perpetual debt used to buy back shares.
- $V_U$, $V_L$: unlevered and levered firm values; $E$: levered equity value; $PV(\text{TS})$: value of the tax shield.

**Step 1 — After-tax cash flow of the all-equity firm.** With no interest, the whole EBIT is taxed:

$$\begin{aligned} EBIT(1 - T_C) &= (92{,}000)(1 - 0.35) \\ &= (92{,}000)(0.65) \\ &= 59{,}800 \text{ per year forever.} \end{aligned}$$

**Step 2 — Unlevered value.** A level perpetuity is worth its cash flow divided by its discount rate:

$$\begin{aligned} V_U &= \frac{EBIT(1-T_C)}{R_U} \\ &= \frac{(59{,}800)}{(0.15)} \\ &= 398{,}666.67. \end{aligned} \tag{3.1}$$

**Step 3 — Value of the tax shield.** The yearly interest is $R_D D = (0.09)(60{,}000) = 5{,}400$, and deducting it saves $T_C$ times that in tax:

$$T_C R_D D = (0.35)(5{,}400) = 1{,}890 \text{ per year.}$$

This perpetuity is as risky as the debt, hence discounted at $R_D$; the $R_D$ cancels:

$$\begin{aligned} PV(\text{TS}) &= \frac{T_C R_D D}{R_D} \\ &= T_C D \\ &= \frac{(1{,}890)}{(0.09)} = (0.35)(60{,}000) \\ &= 21{,}000. \end{aligned} \tag{3.2}$$

**Step 4 — Levered value (MM I with taxes).** Adding (3.2) to (3.1),

$$\begin{aligned} V_L &= V_U + T_C D \\ &= (398{,}666.67) + (21{,}000) \\ &= 419{,}666.67, \\ E &= V_L - D \\ &= (419{,}666.67) - (60{,}000) \\ &= 359{,}666.67. \end{aligned} \tag{3.3}$$

**Answer (firm values).** Unlevered: $V_U = \$398{,}666.67$. With \$60,000 of debt: $V_L = \$419{,}666.67$ (debt \$60,000, equity \$359,666.67).

**Intuition.** The government's claim on EBIT falls by \$1,890 a year once interest is deductible; that perpetual saving, worth $T_C D$, accrues to the firm's investors. The borrowing rate does not affect the answer because it scales both the saving and its discount rate.

**Common mistakes.**

- Capitalising pre-tax EBIT (92,000/0.15) instead of the after-tax cash flow.
- Discounting the tax shield at $R_U$: 1,890/0.15 = 12,600 understates it; with fixed perpetual debt the shield is as safe as the debt.
- Adding the whole \$60,000 to firm value: the debt replaces equity; only $T_C D$ is new value.

**Book answer.** None in the local copy (selected answers are in the web-only Apêndice C).

**Ref:** Ross Cap. 16, §16.4 "Proposições I e II de M&M com impostos sobre o lucro da pessoa jurídica", pp. 551–556.

## 4. APV with constant debt — Higgins 8.23, p. 333

**Statement.**

**Source:** Higgins, Ch. 8, Exercise 8.23, p. 333 (PDF 354) · Answer in book: yes, p. 425

You are valuing a project for Gila Corp. using the APV method. You already found the net present value of free cash flows from the project (discounted at the appropriate cost of equity) to be \$500,000. The only

important side effect of financing is the present value of interest tax shields. The project will be partially financed by a constant $1 million in debt over the life of the project, which is three years. Gila's tax rate is 40 percent, and the current interest rate applicable for the project's debt is 8 percent (assume this rate is also appropriate for discounting the interest tax shields). Assuming tax shields are realized at the end of each year, what is the APV of the project?

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**What is asked.** The adjusted present value of Gila's project: its base-case NPV plus the value of the interest tax shields from $1 million of debt held for three years.

**Approach.** $APV$ values the project as if all-equity financed (given: $500,000) and then adds the PV of each financing side effect; here the only one is the *interest tax shield*. The debt is a fixed amount for three years, so the shield is a three-year annuity, and the statement says to discount it at the 8% debt rate.

**Notation.**

- $NPV_0 = 500{,}000$: base-case NPV of the project's free cash flows.
- $D = 1{,}000{,}000$: constant debt for $T = 3$ years; $r = 0.08$: interest and discount rate of the shields; $t = 0.40$: tax rate.
- $TS$: annual interest tax shield, received at the end of years 1–3; $A$: annuity factor; $APV$: adjusted present value.

**Step 1 — Annual shield.** Interest is charged on the constant balance, and the shield is the tax rate times the interest:

$$\begin{aligned} \text{interest} &= D\,r = (1{,}000{,}000)(0.08) = 80{,}000, \\ TS &= t \cdot D\,r \\ &= (0.40)(80{,}000) \\ &= 32{,}000. \end{aligned} \tag{4.1}$$

**Step 2 — Three-year annuity factor at 8%.** Discount each year's $1 and add:

$$\begin{aligned} A &= \frac{1}{1.08} + \frac{1}{1.08^2} + \frac{1}{1.08^3} \\ &= (0.925926) + (0.857339) + (0.793832) \\ &= 2.577097. \end{aligned} \tag{4.2}$$

The closed form gives the same: $A = \big[1 - 1/1.08^3\big]/0.08 = \big[1 - 1/1.259712\big]/0.08 = 2.577097$.

**Step 3 — PV of the shields and APV.** Multiply the level shield (4.1) by (4.2) and add the base case:

$$\begin{aligned} PV(\text{TS}) &= TS \cdot A \\ &= (32{,}000)(2.577097) \\ &= 82{,}467.10, \end{aligned} \tag{4.3}$$

$$\begin{aligned} APV &= NPV_0 + PV(\text{TS}) \\ &= (500{,}000) + (82{,}467.10) \\ &= 582{,}467.10. \end{aligned} \tag{4.4}$$

**Answer (APV).** $APV = \$582{,}467.10$.

**Intuition.** APV separates the value of the operations ($500,000) from the value created by the financing side effect. With a fixed debt the shields are as safe as the debt, so they are discounted at the debt rate.

**Common mistakes.**

- Treating the shield as perpetual ($tD$ = 400,000, APV \$900,000): the debt lives only three years.
- Adding the interest (\$80,000 a year) instead of the tax saved on it (\$32,000).
- Discounting the shields at the cost of equity, which the statement rules out.

**Book answer.** Matches the book's answer, \$582,467 (p. 425).

**Ref:** Higgins Ch. 8, Appendix "Asset Beta and Adjusted Present Value", pp. 318–325 (APV, p. 321).

## 5. MM II with a rising cost of debt — Berk 16.5, p. 491

**Statement.**

**Source:** Berk, Ch. 16, Exercise 16.5, p. 491 (PDF 522) · Answer in book: no

Hardmon Enterprises is currently an all-equity firm with an expected return of 12%. It is considering borrowing money to buy back some of its existing shares, thus increasing its leverage.

a. Suppose Hardmon borrows to the point that its debt-equity ratio is 0.50. With this amount of debt, the debt cost of capital is 6%. What will the expected return of equity be after this transaction?

b. Suppose instead Hardmon borrows to the point that its debt-equity ratio is 1.50. With this amount of debt, Hardmon's debt will be much riskier. As a result, the debt cost of capital will be 8%. What will the expected return of equity be in this case?

c. A senior manager argues that it is in the best interest of the shareholders to choose the capital structure that leads to the highest expected return for the stock. How would you respond to this argument?

**Context needed to solve it** *From Berk Ch. 16, Problems, section "Capital Structure in Perfect Capital Markets" (instruction that heads this problem section), p. 490 (PDF 521):*

For problems in this section assume no taxes or distress costs.

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**What is asked.** (a) Hardmon's expected equity return if it levers to $D/E = 0.50$ and pays 6% on debt. (b) The same with $D/E = 1.50$ and an 8% debt rate. (c) Whether shareholders are best served by the capital structure with the highest expected stock return.

**Approach.** With no taxes or distress costs (section heading), *MM I* fixes the firm's total value and *MM II* follows: the expected return on the assets (the WACC) stays at the unlevered 12%, so the cost of equity rises with leverage to keep the weighted average at 12%. We derive MM II from that average and apply it, letting $r_D$ change with the debt level.

**Notation.**

- $r_U = 0.12$: expected return on unlevered equity (= on the assets, unchanged by leverage without taxes).
- $D/E$: debt-equity ratio (0.50 or 1.50); $r_D$: debt cost of capital (0.06 or 0.08); $r_E$: expected return on levered equity; $V = D + E$.

**Step 1 — Derive MM II.** The assets' return is the value-weighted average of the returns on the claims that finance them:

$$r_U = \frac{E}{D+E}\, r_E + \frac{D}{D+E}\, r_D.$$

Multiply both sides by $D + E$:

$$(D+E)\, r_U = E\, r_E + D\, r_D.$$

Move $Dr_D$ to the left and write $(D+E)r_U = Er_U + Dr_U$:

$$E\, r_E = E\, r_U + D\, r_U - D\, r_D = E\, r_U + D\,(r_U - r_D).$$

Divide by $E > 0$:

$$r_E = r_U + \frac{D}{E}(r_U - r_D). \tag{5.1}$$

**Step 2 — (a) $D/E = 0.50$, $r_D = 6\%$.** Substituting into (5.1),

$$\begin{aligned} r_E &= (0.12) + (0.50)(0.12 - 0.06) \\ &= (0.12) + (0.50)(0.06) \\ &= (0.12) + (0.03) \\ &= 0.15. \end{aligned} \tag{5.2}$$

**Answer (a).** $r_E = 15\%$.

**Step 3 — (b) $D/E = 1.50$, $r_D = 8\%$.** Substituting into (5.1),

$$\begin{aligned} r_E &= (0.12) + (1.50)(0.12 - 0.08) \\ &= (0.12) + (1.50)(0.04) \\ &= (0.12) + (0.06) \\ &= 0.18. \end{aligned} \tag{5.3}$$

**Answer (b).** $r_E = 18\%$.

**Step 4 — Check: the WACC is unchanged.** Convert $D/E$ into weights: $E/V = 1/(1 + D/E)$ and $D/V = (D/E)/(1 + D/E)$. In (a) $E/V = 1/1.5 = 0.666667$, $D/V = 0.5/1.5 = 0.333333$; in (b) $E/V = 1/2.5 = 0.4$, $D/V = 1.5/2.5 = 0.6$. Then

(a) $(0.666667)(0.15) + (0.333333)(0.06) = 0.10 + 0.02 = 0.12$,
(b) $(0.4)(0.18) + (0.6)(0.08) = 0.072 + 0.048 = 0.12$.

**Step 5 — (c) Reply to the manager.** The higher expected return in (5.3) is pure compensation for higher risk: equity bears the same business risk on a thinner base. By MM I the firm's value, and therefore the share price, is the same at every debt level; the WACC stays at 12% (Step 4). Choosing leverage to maximise $r_E$ does not raise shareholder wealth; shareholders can also lever up on their own (homemade leverage) if they want a riskier, higher-return position.

**Answer (c).** Reject the argument: under MM a higher $r_E$ only compensates for higher risk; firm value and the share price do not change, so shareholders are no better off.

**Intuition.** The cost of equity rises linearly with $D/E$ at slope $r_U - r_D$. When debt becomes riskier ($r_D$ rises), part of the risk moves to creditors and $r_E$ rises less steeply, but the total required return on the assets is still $r_U$.

**Common mistakes.**

- Using $D/V$ instead of $D/E$ in MM II: in (a) that gives $0.12 + (0.333333)(0.06) = 0.14$.
- Keeping $r_D = 6\%$ in (b): $0.12 + (1.50)(0.06) = 0.21$ overstates $r_E$, since riskier debt takes on part of the risk.
- Equating a higher expected return with a higher value: value depends on cash flows and their risk together.

**Book answer.** None in the book.

**Ref:** Berk Ch. 16, §16.2, pp. 464–472 (MM II, leverage and the cost of equity).

## 6. Tax shields on amortising debt — Berk 16.11, p. 492

**Statement.**

**Source:** Berk, Ch. 16, Exercise 16.11, p. 492 (PDF 523) · Answer in book: no

Your firm currently has \$100 million in debt outstanding with a 10% interest rate. The terms of the loan require the firm to repay \$25 million of the balance each year. Suppose that the marginal corporate tax rate is 40%, and that the interest tax shields have the same risk as the loan. What is the present value of the interest tax shields from this debt?

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**What is asked.** The present value of the interest tax shields generated by a \$100 million loan at 10% that is paid down by \$25 million a year, with a 40% tax rate and shields as risky as the loan.

**Approach.** The *interest tax shield* each year is $\tau_c$ times that year's interest, and interest is charged on the balance still outstanding, so the shields shrink as the loan amortises. Because the debt is not permanent, the shortcut $\tau_c D$ does not apply: build the year-by-year schedule and discount each shield at the loan rate, as the statement prescribes.

**Notation.**

- $B_t$: debt outstanding during year $t = 1, \ldots, 4$ (millions); $B_1 = 100$, $B_{t+1} = B_t - 25$.
- $r_D = 0.10$: interest rate and discount rate of the shields; $\tau_c = 0.40$: marginal corporate tax rate.
- $TS_t = \tau_c r_D B_t$: tax shield received at the end of year $t$.

**Assumption.** The first \$25 million repayment is made one year from now, together with the first interest payment, and interest in year $t$ is charged on the balance at the start of the year. The loan is then repaid after four years.

**Step 1 — Schedule.** Apply the 10% rate to each start-of-year balance and the 40% tax rate to each interest payment:

| Year $t$ | Balance $B_t$ | Interest $r_D B_t$ | Repayment | Shield $\tau_c r_D B_t$ |
|---|---|---|---|---|
| 1 | 100 | 10.0 | 25 | 4 |
| 2 | 75 | 7.5 | 25 | 3 |
| 3 | 50 | 5.0 | 25 | 2 |
| 4 | 25 | 2.5 | 25 | 1 |

$$TS_t = (0.40)(0.10)B_t = (0.04)B_t \in \{4,\ 3,\ 2,\ 1\}. \tag{6.1}$$

**Step 2 — Discount factors at 10%.** $1/1.10 = 0.909091$, $1/1.10^2 = 1/1.21 = 0.826446$, $1/1.10^3 = 1/1.331 = 0.751315$, $1/1.10^4 = 1/1.4641 = 0.683013$.

**Step 3 — Present value.** Multiply each shield (6.1) by its factor and add:

$$\begin{aligned} PV(\text{TS}) &= \sum_{t=1}^{4} \frac{TS_t}{(1+r_D)^t} \\ &= \frac{(4)}{1.10} + \frac{(3)}{1.21} + \frac{(2)}{1.331} + \frac{(1)}{1.4641} \\ &= 3.636364 + 2.479339 + 1.502630 + 0.683013 \\ &= 8.301346. \end{aligned} \tag{6.2}$$

**Answer (PV of the shields).** $PV(\text{TS}) \approx \$8.30$ million.

**Intuition.** Each year's shield is proportional to the debt still outstanding, so amortisation shrinks the shield stream. A permanent \$100 million loan would be worth $\tau_c D = \$40$ million in shields; repaying it over four years leaves $8.301346/40 = 0.2075$, about a fifth of that.

**Common mistakes.**

- Using $\tau_c D = \$40$ million, the value for *permanent* debt.
- Charging interest on the end-of-year balance (shields 3, 2, 1, 0), which gives $3/1.1 + 2/1.21 + 1/1.331 = 5.131480$ and misses the first year's interest.
- Adding the shields undiscounted ($10 million) or discounting them at an unlevered rate.

**Book answer.** None in the book.

**Ref:** Berk Ch. 16, §16.3 "Debt and Taxes", pp. 473–479 (value of the interest tax shield).

## 7. Homemade leverage between twin firms — Ross 16.18, p. 578

**Statement.**

**Source:** Ross, Cap. 16, Exercise 16.18, p. 578 (PDF 602) · Answer in book: no (selected answers in the web-only Apêndice C)

As companhias Weber e Cavaleiro são idênticas em todos os aspectos, exceto que a Weber não é alavancada. As informações financeiras das duas empresas são mostradas no quadro a seguir. Todos os grupos de lucros são perpetuidades, e nenhuma das empresas paga impostos. As duas empresas distribuem imediatamente todos os lucros disponíveis aos acionistas.

| | Weber | Cavaleiro |
|---|---|---|
| Lucro operacional projetado | \$400.000 | \$400.000 |
| Juros sobre a dívida ao fim do ano | — | \$72.000 |
| Valor de mercado do capital próprio | \$2.500.000 | \$1.632.000 |
| Valor de mercado da dívida | — | \$1.200.000 |

a. Um investidor que pode tomar emprestado a 6% ao ano deseja comprar 5% do capital da Cavaleiro. Ele poderia aumentar seu retorno comprando 5% do capital da Weber se tomar emprestado de tal forma que os custos líquidos das estratégias sejam iguais?

b. Dadas as duas estratégias de investimentos de (a), qual será a estratégia escolhida pelos investidores? Quando esse processo terminará?

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**What is asked.** (a) An investor who can borrow at 6% wants 5% of Cavaleiro (levered). Can she earn more by buying 5% of Weber (unlevered) and borrowing personally, spending the same amount of her own money? (b) Which strategy investors will choose, and when this process stops.

**Approach.** This is *homemade leverage*, the arbitrage behind *MM Proposition I* without taxes. Two firms with the same operating income must have the same total value; if not, investors replicate one firm's equity with the other firm's equity plus personal borrowing or lending. Here $V_C = 1{,}632{,}000 + 1{,}200{,}000 = 2{,}832{,}000 > V_W = 2{,}500{,}000$, so Cavaleiro is overpriced and the cheaper route to its payoff goes through Weber.

**Notation.**

- $X$: perpetual operating income of each firm (expected 400,000; random, identical for both); no taxes.
- Weber (unlevered): $E_W = V_W = 2{,}500{,}000$. Cavaleiro (levered): $E_C = 1{,}632{,}000$, $D_C = 1{,}200{,}000$, interest $I_C = 72{,}000$ per year, so $r_D = 72{,}000/1{,}200{,}000 = 0.06$.
- $\alpha = 0.05$: fraction bought; $r = 0.06$: personal borrowing rate; $L$: amount borrowed personally.

**Assumption.** "Os custos líquidos das estratégias sejam iguais" is read as: both strategies require the same own funds today, the \$81,600 that 5% of Cavaleiro costs.

**Step 1 — Strategy C: 5% of Cavaleiro.** Cavaleiro's shareholders receive operating income minus interest, $X - I_C$; with the projected income that is $400{,}000 - 72{,}000 = 328{,}000$. Then

$$\text{cost}_C = \alpha E_C = (0.05)(1{,}632{,}000) = 81{,}600, \tag{7.1}$$

$$\text{income}_C = \alpha(X - I_C) = (0.05)(328{,}000) = 16{,}400, \tag{7.2}$$

a return of 16,400/81,600 = 20.10%.

**Step 2 — Strategy W: 5% of Weber with the same own funds.** The Weber stake costs (0.05)(2,500,000) = 125,000 and earns (0.05)(400,000) = 20,000. With own funds (7.1), the rest is borrowed at 6%:

$$L = 125{,}000 - 81{,}600 = 43{,}400, \tag{7.3}$$

$$\begin{aligned} \text{income}_W &= 20{,}000 - rL \\ &= (20{,}000) - (0.06)(43{,}400) \\ &= (20{,}000) - (2{,}604) = 17{,}396, \end{aligned} \tag{7.4}$$

a return of 17,396/81,600 = 21.32%.

**Step 3 — (a) Same risk, higher income.** Write both incomes for any realisation of $X$:

$$\begin{aligned} \text{income}_C &= 0.05\,(X - 72{,}000) = 0.05X - 3{,}600, \\ \text{income}_W &= 0.05X - (0.06)(43{,}400) = 0.05X - 2{,}604. \end{aligned}$$

Both move one-for-one with $0.05X$, so they carry the same operating risk, and W pays $3{,}600 - 2{,}604 = 996$ more in every state. The W position even uses less leverage: \$43,400 of personal debt against her 5% share of Cavaleiro's \$1,200,000, i.e. \$60,000.

**Answer (a).** Yes: for the same \$81,600, 5% of Weber plus \$43,400 borrowed pays \$17,396 (21.32%) against \$16,400 (20.10%), \$996 more in every state at no extra risk.

**Step 4 — The underlying arbitrage.** Replicating strategy C exactly needs the investor's share of Cavaleiro's debt, $L = \alpha D_C = (0.05)(1{,}200{,}000) = 60{,}000$: income $20{,}000 - (0.06)(60{,}000) = 20{,}000 - 3{,}600 = 16{,}400$, identical to (7.2), for own funds $125{,}000 - 60{,}000 = 65{,}000$. The saving is

$$81{,}600 - 65{,}000 = 16{,}600 = \alpha(V_C - V_W) = (0.05)(2{,}832{,}000 - 2{,}500{,}000) = (0.05)(332{,}000), \tag{7.5}$$

and investing it at 6% yields (0.06)(16,600) = 996 a year, the gap found in Step 3.

**Step 5 — (b) What investors do and when it stops.** Every Cavaleiro shareholder gains by selling Cavaleiro shares and buying Weber shares with personal borrowing. Selling pressure lowers $E_C$ and buying pressure raises $E_W$ until the arbitrage profit (7.5) is zero, i.e. until $V_C = V_W$ (MM I). If Weber's value stays at \$2.5 million, it stops when $E_C = 2{,}500{,}000 - 1{,}200{,}000 = 1{,}300{,}000$. Cavaleiro's cost of equity is then 328,000/1,300,000 = 25.23%, which is MM II with $r_U = 400{,}000/2{,}500{,}000 = 0.16$:

$$r_E = 0.16 + \tfrac{1.2}{1.3}(0.16 - 0.06) = 0.16 + (0.923077)(0.10) = 0.16 + 0.092308 = 0.252308.$$

**Answer (b).** Investors sell Cavaleiro and buy Weber on personal margin; the process ends when the two firms have the same total value, $V_C = V_W$ (e.g. $E_C = \$1{,}300{,}000$ if $V_W$ stays at \$2.5 million).

**Intuition.** Investors can borrow on the same terms as the firm, so corporate leverage adds nothing they cannot build themselves. A levered firm that trades at a premium is selling something investors can make more cheaply at home, and arbitrage removes the premium.

**Common mistakes.**

- Comparing the firms' returns on equity without equalising own funds (or risk): Weber's unlevered 16% is not comparable with Cavaleiro's 20.10%.
- Borrowing the full \$60,000 and then comparing incomes: that equalises income, not cost; the gain then shows up as \$16,600 saved today instead of \$996 a year.
- Forgetting that the process stops at equal *firm* values, not equal share prices or equal returns.

**Book answer.** None in the local copy (selected answers are in the web-only Apêndice C).

**Ref:** Ross Cap. 16, §16.2 "O efeito da alavancagem financeira" (homemade leverage), pp. 542–547; §16.3, pp. 547–551 (Propositions I and II).

# 8. Announcement effect of a debt-financed repurchase with distress costs — Berk 16.22, p. 494

**Statement.**

**Source:** Berk, Ch. 16, Exercise 16.22, p. 494 (PDF 525) · Answer in book: no

Hawar International is a shipping firm with a current share price of \$5.50 and 10 million shares outstanding. Suppose that Hawar announces plans to lower its corporate taxes by borrowing \$20 million and repurchasing shares, that Hawar pays a corporate tax rate of 30%, and that shareholders expect the change in debt to be permanent.

a. If the only imperfection is corporate taxes, what will the share price be after this announcement?

b. Suppose the only imperfections are corporate taxes and financial distress costs. If the share price rises to \$5.75 after this announcement, what is the PV of financial distress costs Hawar will incur as the result of this new debt?

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**What is asked.** (a) The share price after Hawar announces a permanent \$20 million debt-financed buyback, if corporate taxes are the only imperfection. (b) If the price actually rises only to \$5.75, the PV of the financial distress costs the market attributes to the new debt.

**Approach.** Permanent debt creates an interest tax shield worth $\tau_c D$ (*MM I with taxes*). When the plan is announced, the whole change in firm value is capitalised into the price of the existing shares, before any shares are bought back. With distress costs, the *static trade-off* identity $V_L = V_U + \tau_c D - PV(\text{FDC})$ lets us back out $PV(\text{FDC})$ from the observed price.

**Notation.**

- $P_0 = 5.50$, $N = 10$ million shares, so $E_0 = (5.50)(10) = 55$ million.
- $D = 20$ million of permanent new debt; $\tau_c = 0.30$; $P_1$: price after the announcement; $PV(\text{FDC})$: PV of financial distress costs.

**Assumption.** The announcement is credible and all the value change is priced at once; shares are later bought back at the post-announcement price, so the repurchase itself transfers no value.

**Step 1 — (a) Value of the tax shield.** Permanent debt adds (cf. (3.2))

$$\tau_c D = (0.30)(20) = 6 \text{ million}.$$

**Step 2 — (a) Price after the announcement.** The \$6 million goes to the 10 million shares outstanding at the announcement:

$$\begin{aligned} P_1 &= \frac{E_0 + \tau_c D}{N} \\ &= \frac{(55) + (6)}{(10)} \\ &= 6.10. \end{aligned} \tag{8.1}$$

Check that the buyback leaves it there: \$20 million buys $20/6.10 = 3.278689$ million shares; equity falls to $61 - 20 = 41$ million over $10 - 3.278689 = 6.721311$ million shares, i.e. $41/6.721311 = 6.10$.

**Answer (a).** $P_1 = \$6.10$ (a gain of \$0.60 per share).

**Step 3 — (b) Distress costs implied by \$5.75.** The observed gain in equity value is

$$(5.75)(10) - (55) = 57.5 - 55 = 2.5 \text{ million}.$$

By the trade-off identity this gain equals the tax benefit net of distress costs; solving for the distress costs:

$$\begin{aligned} 2.5 &= \tau_c D - PV(\text{FDC}) \\ PV(\text{FDC}) &= \tau_c D - 2.5 \\ &= (6) - (2.5) \\ &= 3.5 \text{ million.} \end{aligned} \tag{8.2}$$

(The buyback at \$5.75 again leaves the price unchanged: $20/5.75 = 3.478261$ million shares, $37.5/(10 - 3.478261) = 37.5/6.521739 = 5.75$.)

**Answer (b).** $PV(\text{FDC}) = \$3.5$ million.

**Intuition.** The market pays the shareholders for the expected tax savings and charges them for expected distress costs. The smaller price jump (\$0.25 against \$0.60) shows that the market expects distress costs to eat more than half of the tax benefit of this debt.

**Common mistakes.**

- Dividing the gain by the post-repurchase share count: the announcement gain accrues to all 10 million shares outstanding when the news arrives.
- Looking for an interest rate: with permanent debt the shield is worth $\tau_c D$ whatever the rate.
- Reading \$5.75 as a loss because it is below \$6.10: the \$3.5 million is the gap between the two prices times 10 million shares.

**Book answer.** None in the book.

**Ref:** Berk Ch. 16, §16.3 pp. 473–479 and §16.5 "Optimal Capital Structure: The Tradeoff Theory", pp. 480–482.

## 9. After-tax WACC with riskless perpetual debt, then a debt-for-equity exchange — Grinblatt & Titman 13.15–13.16, p. 494

**Statement.**

**Source:** Grinblatt & Titman, Ch. 13, Exercises 13.15 and 13.16, p. 494 (PDF 507) · Answer in book: no

**[13.15]** The Akron Company consists of \$50 million in perpetual riskless debt and \$50 million in equity. The current market value of its assets is \$100 million and the beta of its equity return is 1.2. Assume the risk-free rate is 8 percent, the expected return of the market portfolio is 13 percent per year, and the CAPM is true. Compute the expected return of Akron's equity and its WACC assuming a 40 percent corporate tax rate.

**[13.16]** Akron, from the last example, is considering an exchange offer where half of Akron's outstanding debt (\$25 million) is retired. The purchase of this debt would be financed by issuing \$25 million in equity to the debt holders of Akron. Assuming debt policy that is consistent with the Hamada model, what will Akron's new WACC be after the exchange offer?

**Context needed to solve it** *From Grinblatt & Titman Ch. 13, Eq. (13.8), p. 476 (PDF 489):*

$$\text{WACC} = w_E \bar{r}_E + w_D (1 - T_c) \bar{r}_D \tag{13.8}$$

where

- WACC = weighted average cost of capital
- $w_E = \dfrac{E}{D+E}$ = market value of equity over market value of all financing
- $w_D = \dfrac{D}{D+E}$ = market value of debt over market value of all financing
- $T_c$ = marginal corporate tax rate if interest is fully tax deductible (or, more generally, the debt financing subsidy in percentage terms)

The costs of the financing components are

- $\bar{r}_E$ = the expected return on equity to investors

- $\bar{r}_D$ = the expected return on debt to investors

*From Grinblatt & Titman Ch. 13, Exhibit 13.1 and "Static Perpetual Risk-Free Debt", pp. 464–465 (PDF 477–478):*

Exhibit 13.1 — Balance Sheet for a Firm with Leverage When Debt Interest Is Corporate Tax Deductible

| Assets | | Liabilities and Equity | |
|---|---|---|---|
| Debt Tax Shield ($TX$) | $T_cD$ | Debt | $D$ |
| Unlevered Assets ($UA$) | $D+E-T_cD$ | Equity | $E$ |

Viewing the assets of the firm with value $A$ as a portfolio of unlevered assets with value $UA$ and debt tax shields with value $TX$ implies that the beta (or expected return) of the assets is the portfolio-weighted average of the betas (or expected returns) of the unlevered assets and debt tax shields; that is

$$\beta_A = \left(\frac{UA}{D+E}\right)\beta_{UA} + \left(\frac{TX}{D+E}\right)\beta_{TX} \qquad (13.1a)$$

and

$$\bar{r}_A = \left(\frac{UA}{D+E}\right)\bar{r}_{UA} + \left(\frac{TX}{D+E}\right)\bar{r}_{TX} \qquad (13.1b)$$

where $\beta_{UA}$ = beta risk of the unlevered assets, $\beta_{TX}$ = beta risk of the debt tax shield, $\bar{r}_{UA}$ = expected return of the unlevered assets, $\bar{r}_{TX}$ = expected return of the debt tax shield.

**Static Perpetual Risk-Free Debt.** Let $T_c$ denote the effective corporate tax rate. Exhibit 13.1 has the firm's debt tax shield expressed as

$$TX = T_cD \qquad (13.2)$$

and thus

$$UA = D + E - T_cD \qquad (13.3)$$

The values for $TX$ and $UA$ both in Exhibit 13.1 and in equations (13.2) and (13.3) are developed in a model by Hamada (1972). If the firm issues default-free perpetual debt (that is, debt that never matures) with aggregate face amount of $D$ and interest payments equal to the risk-free rate, each interest payment of $Dr_f$ saves $T_cDr_f$ in taxes in the year it is paid. The present value of the tax savings from the interest payments [. . . ] is

$$T_cD = \frac{T_cDr_f}{r_f}$$

The same present value is also achieved when the risk-free rate changes over time and the firm rolls over one-period risk-free debt.

The Hamada model assumes that the tax shield is riskless and thus, each period's tax deduction arising from an interest payment should be discounted back to date 0 at the risk-free rate. This implies that the beta of the debt tax shield in the Hamada model is zero. Using this observation, and substituting equations (13.2) and (13.3) into equation (13.1a) yields

$$\beta_A = \left[\frac{D+E-T_cD}{D+E}\right]\beta_{UA} \qquad (13.4)$$

[. . . ] The key assumptions that lead to this result are: (1) The debt is perpetual; in other words, it is either a perpetuity or consists of rolled over short-term debt positions. (2) The debt is default-free and pays the risk-free rate. (3) The face value of the debt and the tax rate do not change over time. The third assumption distinguishes the Hamada model from other models that will be discussed later in this chapter.

**Equity Betas and Asset Betas.** Equity betas are also affected by taxes. Chapter 11 indicated that (assuming risk-free debt) the equity beta is

$$\beta_E = \left(1 + \frac{D}{E}\right)\beta_A \qquad (13.5)$$

[. . . ] With a little simplification, this equation, which is based on the Hamada model, becomes

$$\beta_E = \left[1 + (1 - T_c)\frac{D}{E}\right] \beta_{UA} \qquad (13.6)$$

*From Grinblatt & Titman Ch. 13, Eq. (13.11), p. 481 (PDF 494):*

$$\text{WACC} = \bar{r}_A - \left(\frac{D}{D+E}\right) T_c \, \bar{r}_D \qquad (13.11)$$

*From Grinblatt & Titman Ch. 13, "The Adjusted Cost of Capital Formula", p. 482 (PDF 495):*

In the Hamada model, with static perpetual debt, both debt and its tax shield are risk-free, implying $\bar{r}_D = r_f$, $\beta_{TX} = 0$, and $TX = T_c D$. In this case, the portfolio-weighted average of the expected returns of the unlevered assets and the debt tax shield, given in equation (13.1b), simplifies to

$$\bar{r}_A = \left(1 - \frac{T_c D}{D+E}\right) \bar{r}_{UA} + \frac{T_c D}{D+E} \bar{r}_D$$

If we substitute this equation into equation (13.11), equation (13.11) reduces to Modigliani and Miller's (1963) adjusted cost of capital formula, which gives the firm's WACC as a function of its debt to value ratio:

$$\text{WACC} = \bar{r}_{UA} \left[1 - T_c \frac{D}{D+E}\right]$$

---

**What is asked.** 13.15: Akron's expected equity return (from the CAPM) and its after-tax WACC with \$50 million of riskless perpetual debt and a 40% tax rate. 13.16: Akron's WACC after it retires half its debt by issuing \$25 million of equity to the debt holders, under the Hamada model's debt policy.

**Approach.** The CAPM gives $r_E$ from $\beta_E$; G&T's (13.8) then gives the after-tax WACC. For 13.16 the *Hamada model* (static perpetual riskless debt) says the firm is unlevered assets plus a riskless tax shield $T_c D$. The unlevered assets, their beta and their return $r_{UA}$ do not change with leverage, so we unlever once and then use the *MM adjusted cost of capital* $WACC = r_{UA}(1 - T_c D/V)$ at the new debt level, cross-checking through the relevered cost of equity.

**Notation.**

- $D = 50$, $E = 50$ (millions): perpetual riskless debt and equity; $A = D + E = 100$: market value of the assets.
- $\beta_E = 1.2$; $r_f = 0.08$; $\mathbb{E}[r_M] = 0.13$; CAPM holds; $r_D = r_f$ (riskless debt); $T_c = 0.40$.
- $UA$, $TX$: unlevered assets and debt tax shield, $A = UA + TX$; $\beta_{UA}$, $r_{UA}$: their beta and expected return.
- Primes ($D'$, $E'$, $V'$, $WACC'$) denote values after the exchange offer.

**Step 1 — 13.15: cost of equity (CAPM).** The market risk premium is $0.13 - 0.08 = 0.05$, so

$$\begin{aligned} r_E &= r_f + \beta_E(\mathbb{E}[r_M] - r_f) \\ &= (0.08) + (1.2)(0.13 - 0.08) \\ &= (0.08) + (1.2)(0.05) \\ &= (0.08) + (0.06) \\ &= 0.14. \end{aligned} \qquad (9.1)$$

**Step 2 — 13.15: WACC.** Use G&T's after-tax WACC (13.8) with market weights $50/100 = 0.5$ each and (9.1):

$$\begin{aligned} WACC &= \frac{E}{D+E} r_E + \frac{D}{D+E} r_D(1 - T_c) \qquad (9.2) \\ &= (0.5)(0.14) + (0.5)(0.08)(1 - 0.40) \\ &= (0.07) + (0.5)(0.08)(0.60) \\ &= (0.07) + (0.024) \\ &= 0.094. \end{aligned}$$

**Answer (13.15).** $r_E = 14\%$; $WACC = 9.40\%$.

**Step 3 — Unlevered cost of capital (Hamada).** With static perpetual riskless debt, $TX = T_c D$ (13.2) and $UA = D + E - T_c D = 100 - (0.40)(50) = 100 - 20 = 80$ (13.3). Invert (13.6), $\beta_E = \big[1 + (1 - T_c)\frac{D}{E}\big]\beta_{UA}$, by dividing both sides by the bracket:

$$\begin{aligned}
\beta_{UA} &= \frac{\beta_E}{1 + (1 - T_c)D/E} \qquad (9.3)\\
&= \frac{(1.2)}{1 + (1 - 0.40)(50/50)}\\
&= \frac{1.2}{1 + 0.6} = \frac{1.2}{1.6}\\
&= 0.75,\\
r_{UA} &= r_f + \beta_{UA}(\mathbb{E}[r_M] - r_f) \qquad (9.4)\\
&= (0.08) + (0.75)(0.05)\\
&= (0.08) + (0.0375)\\
&= 0.1175.
\end{aligned}$$

**Step 4 — The adjusted cost of capital formula.** Start from (13.11), $WACC = \bar{r}_A - \frac{D}{V} T_c r_D$, and substitute the Hamada form of (13.1b), $\bar{r}_A = (1 - \frac{T_c D}{V}) r_{UA} + \frac{T_c D}{V} r_D$:

$$\begin{aligned}
WACC &= \Big(1 - \frac{T_c D}{V}\Big) r_{UA} + \frac{T_c D}{V} r_D - \frac{D}{V} T_c r_D\\
&= \Big(1 - \frac{T_c D}{V}\Big) r_{UA}\\
&= r_{UA}\Big(1 - T_c \frac{D}{V}\Big), \qquad (9.5)
\end{aligned}$$

since the two $r_D$ terms are equal and cancel. Check at today's leverage: $(0.1175)\big(1 - (0.40)(0.5)\big) = (0.1175)(0.8) = 0.094$, equal to (9.2).

**Step 5 — 13.16: values after retiring \$25 million of debt.** The unlevered assets do not change; the shield shrinks to $T_c D' = (0.40)(25) = 10$. Hence

$$\begin{aligned}
V' &= UA + T_c D' = (80) + (10) = 90, \qquad (9.6)\\
E' &= V' - D' = (90) - (25) = 65,
\end{aligned}$$

of which \$25 million is the new shares given to the old creditors and $65 - 25 = 50 - (0.40)(25) = 40$ is the old shareholders' stake: they bear the \$10 million loss of shield value.

**Step 6 — 13.16: new WACC.** The new debt ratio is $D'/V' = 25/90 = 0.277778$; substitute it into (9.5) with (9.4):

$$\begin{aligned}
WACC' &= r_{UA}\Big(1 - T_c \frac{D'}{V'}\Big) \qquad (9.7)\\
&= (0.1175)\big(1 - (0.40)(0.277778)\big)\\
&= (0.1175)(0.888889)\\
&= 0.104444.
\end{aligned}$$

**Step 7 — Cross-check through the cost of equity.** Relever with (13.6) at $D'/E' = 25/65 =$

0.384615:

$$\begin{aligned}\beta'_E &= \beta_{UA}\Big[1 + (1 - T_c)\frac{D'}{E'}\Big] \\ &= (0.75)\big[1 + (0.6)(0.384615)\big] = (0.75)(1 + 0.230769) \\ &= (0.75)(1.230769) = 0.923077, \\ r'_E &= (0.08) + (0.923077)(0.05) = (0.08) + (0.046154) = 0.126154, \\ WACC' &= \tfrac{65}{90}(0.126154) + \tfrac{25}{90}(0.08)(0.6) \\ &= (0.091111) + (0.013333) = 0.104444,\end{aligned}$$

the same as (9.7). The WACC rises from 9.40% to 10.44%.

**Answer (13.16).** $WACC' = 10.44\%$.

**Assumption.** The debt ratio in (9.7) is measured at post-exchange market values. Using $D'/V = 25/100$ (ignoring the lost shield) would give $(0.1175)(1 - 0.40 \cdot 0.25) = 10.575\%$; that mixes the old firm value with the new debt and is not consistent with the Hamada model.

**Intuition.** In the Hamada model the only thing debt adds is a riskless tax shield. Cutting the debt removes part of that cheap, government-funded slice of the financing, so the WACC moves back toward the unlevered cost of capital, 11.75%.

**Common mistakes.**

- Computing the pre-tax WACC, $(0.5)(0.14) + (0.5)(0.08) = 0.11$, instead of (13.8) with $(1 - T_c)$ on debt.
- Keeping $r_E = 14\%$ after the exchange: $\frac{65}{90}(0.14) + \frac{25}{90}(0.048) = 0.114444$ ignores that less leverage makes equity safer.
- Weighting by the old value $V = 100$ (10.575%): retiring debt destroys \$10 million of shield, so $V' = 90$.

**Book answer.** None in the book.

**Ref:** Grinblatt & Titman Ch. 13: Hamada model, eqs. (13.1)–(13.7), pp. 463–466; WACC (13.8), p. 476; adjusted cost of capital formula, pp. 481–482.

## 10. Taxes vs. bankruptcy costs in two states — Grinblatt & Titman 16.2, p. 591

**Statement.**

**Source:** Grinblatt & Titman, Ch. 16, Exercise 16.2, p. 591 (PDF 604) · Answer in book: no

Nigel decides he can make zippers at night for one period and will have cash flows next period of \$210 if the economy is favorable, and \$66 if the economy is unfavorable. One-third of these proceeds must be paid out in taxes if the firm is all equity financed; however, because of the tax advantage of debt, Nigel saves \$0.05 in taxes for every \$1.00 of debt financing that he uses. Assume investors are risk neutral, the riskless rate is 10 percent per period, and the probability of each state is .5. Also assume that if Nigel's firm goes bankrupt and debt holders take over, the legal fees and other bankruptcy costs total \$20.

a. If Nigel organizes his firm as all equity, what will it be worth?

b. Suppose Nigel's firm sold a zero-coupon bond worth \$44 at maturity next period. How much would the firm receive for the debt?

c. With the debt level above, how much would the equity be worth?

d. How much would the firm be worth?

e. Would the firm be worth more if it had a debt obligation of \$70 next period?

**What is asked.** (a) The value of Nigel's firm if all equity financed. (b) What the firm raises today by selling a zero-coupon bond that promises \$44 next period. (c) The value of the equity with that bond outstanding. (d) The total value of the firm with that bond. (e) Whether the firm would be worth more with a \$70 obligation instead.

**Approach.** Investors are risk neutral, so every claim is worth its expected payoff discounted at the 10% riskless rate. Debt brings a tax saving of \$0.05 per dollar raised; debt large enough to default in the bad state also triggers a \$20 bankruptcy cost. This is the *static trade-off*: firm value $= V_U + PV(\text{tax saving}) - PV(\text{bankruptcy costs})$. For each debt level we first check whether default happens, then price debt and equity state by state.

**Notation.**

- $C_G = 210$, $C_B = 66$: pre-tax cash flows next period in the favourable and unfavourable state, each with probability $\frac{1}{2}$.
- $t = \frac{1}{3}$: tax rate of the all-equity firm; $s = 0.05$: tax saved per dollar of debt financing; $r_f = 0.10$; investors are risk neutral.
- $F$: face value of the zero-coupon debt; $D$: its market value today; $K = 20$: bankruptcy cost; $V_U$, $E$, $V$: unlevered value, equity, levered firm value.

**Assumption.** "Debt financing used" is the amount raised today, $D$; the saving $sD$ is realised next period in both states (if the firm defaults it accrues to the creditors who take it over). The answer to (e) does not depend on this reading (Step 7).

**Step 1 — (a) All-equity value.** After paying one third in tax, the firm keeps two thirds: $210(1 - \frac{1}{3}) = 140$ and $66(1 - \frac{1}{3}) = 44$. Discount the expected value:

$$\begin{aligned} V_U &= \frac{\frac{1}{2}(140) + \frac{1}{2}(44)}{1.10} \\ &= \frac{70 + 22}{1.10} = \frac{92}{1.10} = \frac{920}{11} \\ &= 83.6364. \end{aligned} \tag{10.1}$$

**Answer (a).** $V_U = \$83.64$.

**Step 2 — (b) Debt with $F = 44$.** Guess that the debt is safe; then it is worth its face discounted at the riskless rate, and the tax saving follows:

$$D = \frac{44}{1.10} = 40, \qquad sD = (0.05)(40) = 2.$$

Verify the guess: in the bad state the firm has $44 + 2 = 46 \geq 44$, so it pays in full in both states.

**Answer (b).** The firm receives $D = \$40$.

**Step 3 — (c) Equity.** Equity gets what is left after paying 44: $140 + 2 - 44 = 98$ in the good state and $46 - 44 = 2$ in the bad state:

$$\begin{aligned} E &= \frac{\frac{1}{2}(98) + \frac{1}{2}(2)}{1.10} \\ &= \frac{49 + 1}{1.10} = \frac{50}{1.10} = \frac{500}{11} \\ &= 45.4545. \end{aligned} \tag{10.2}$$

**Answer (c).** $E = \$45.45$.

**Step 4 — (d) Firm value.** Add the two claims:

$$\begin{aligned} V &= D + E \\ &= (40) + (45.4545) \\ &= 85.4545, \end{aligned} \tag{10.3}$$

equal to $V_U + \frac{2}{1.10} = \frac{920}{11} + \frac{20}{11} = \frac{940}{11}$: the PV of the tax saving, 1.8182, is the whole gain.

**Answer (d).** $V = \$85.45$.

**Step 5 — (e) Debt with $F = 70$: price the risky debt.** Now the bad-state cash, $44 + sD$, is below 70 for any plausible $D$, so the firm defaults; creditors take over, pay the \$20 cost and receive $44 + 0.05D - 20 = 24 + 0.05D$. In the good state they get 70. The debt's price solves

$$\begin{aligned} D &= \frac{\frac{1}{2}(70) + \frac{1}{2}(24 + 0.05D)}{1.10} \\ 1.10\,D &= 35 + 12 + 0.025\,D \\ 1.10\,D - 0.025\,D &= 47 \\ 1.075\,D &= 47 \\ D &= \frac{47}{1.075} = 43.72093. \end{aligned} \tag{10.4}$$

Then $sD = (0.05)(43.72093) = 2.18605$ and the bad-state recovery is $24 + 2.18605 = 26.18605 < 70$: default confirmed.

**Step 6 — (e) Equity and firm value.** Equity gets $140 + 2.18605 - 70 = 72.18605$ in the good state and 0 in the bad state:

$$\begin{aligned} E &= \frac{\frac{1}{2}(72.18605) + \frac{1}{2}(0)}{1.10} \\ &= 32.81184, \\ V &= D + E \\ &= (43.72093) + (32.81184) \\ &= 76.53277. \end{aligned} \tag{10.5}$$

Equivalently $V = V_U + PV(sD) - PV(K) = (83.63636) + (1.98732) - (9.09091) = 76.53277$, using $2.18605/1.10 = 1.98732$ and $\frac{1}{2}(20)/1.10 = 9.09091$.

**Step 7 — Robustness.** Whatever the basis of the saving, it is at most $0.05 \times 70$ next period, worth 3.18182, while expected bankruptcy costs are worth 9.09091; so $V \le (83.63636) + (3.18182) - (9.09091) = 77.72727 < 85.4545$ under any reading.

**Answer (e).** No: with a \$70 obligation the firm is worth \$76.53, less than \$85.45.

**Intuition.** This is the static trade-off in miniature. Safe debt adds a small tax subsidy at no cost. Debt large enough to default in the bad state buys a little more subsidy but triggers a deadweight loss of \$20 with probability one half, which dominates.

**Common mistakes.**

- Ignoring corporate tax in (a): $(105 + 33)/1.10 = 125.4545$ is the pre-tax value.
- Pricing the \$70 bond as safe, $70/1.10 = 63.64$: the bad state cannot pay it.
- Thinking the bankruptcy cost hurts only creditors: they price it into $D$, so the shareholders who issue the debt bear it.

**Book answer.** None in the book.

**Ref:** Grinblatt & Titman Ch. 16, §16.1 "Bankruptcy" (direct bankruptcy costs vs. the debt tax advantage), pp. 559–561.

## 11. Risky debt in a binomial market and new debt of equal seniority — Grinblatt & Titman 14.2–14.3, pp. 526–527

**Statement.**

**Source:** Grinblatt & Titman, Ch. 14, Exercise 14.3 (with Exercise 14.2, on which it builds, as part 0), pp. 526–527 (PDF 539) · Answer in book: no

**[14.2]** Consider a single period binomial setting where the riskless interest rate is zero, and there are no taxes. A firm consists of a machine that will produce cash flows of \$210 if the economy is good and \$80 if the economy is bad. The good and bad states occur with equal risk-neutral probability. Initially, the firm has 100 shares outstanding and debt with a face value of \$50 due at the end of the period. What is the share price of the firm?

**[14.3]** Suppose the firm in exercise 14.2 unexpectedly announces that it will issue additional debt, with the same seniority as existing debt and a face value of \$50. The firm will use the entire proceeds to repurchase some of the outstanding shares.

a. What is the market price of the new debt?

b. Just after the announcement, what will the price of a share jump to?

c. Show how a shareholder with 20 percent of the shares outstanding is better off as a result of this transaction when he or she undoes the leverage change.

d. Show how the Modigliani-Miller Theorem still holds.

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**What is asked.** 14.2: The share price of a firm worth \$210 or \$80 next period with \$50 of debt and 100 shares. 14.3: After a surprise issue of \$50 more face of equal-seniority debt used to buy back shares: (a) the price of the new debt; (b) the share price right after the announcement; (c) show that a 20% shareholder who undoes the leverage change ends up better off; (d) show that MM still holds.

**Approach.** With a zero interest rate and equal risk-neutral probabilities, every claim is worth the plain average of its two payoffs. Equal seniority means that in default the cash is shared pro rata by face value, so the new issue makes the old debt risky. *MM Proposition I* says total firm value does not depend on the split; any gain to shareholders must therefore be a transfer from the old creditors. Undoing leverage is *homemade leverage* run in reverse: hold debt alongside the shares.

**Notation.**

- $C_G = 210$, $C_B = 80$: firm cash flow in the good/bad state; risk-neutral probability $\frac{1}{2}$ each; $r_f = 0$; no taxes.
- $F_0 = 50$: face of the existing debt; $F_1 = 50$: face of the new equal-seniority debt; 100 shares initially.
- $V$: firm (machine) value; $D_0$, $D_1$: market values of old and new debt; $P$: share price; $n_R$: shares repurchased.

**Step 1 — 14.2: share price.** With $r_f = 0$, values are risk-neutral expectations:

$$V = \tfrac{1}{2}(210) + \tfrac{1}{2}(80) = 105 + 40 = 145. \tag{11.1}$$

The old debt is safe ($80 \geq 50$), so $D_0 = 50$; equity pays $(210 - 50,\ 80 - 50) = (160,\ 30)$ and is worth $145 - 50 = 95$, so $P = 95/100 = 0.95$.

**Answer (14.2).** $P = \$0.95$.

**Step 2 — (a) Price of the new debt.** Total face becomes $F_0 + F_1 = 100 > 80$, so in the bad state the \$80 is shared pro rata: each issue gets $\frac{50}{100}(80) = 40$. The new debt pays $(50,\ 40)$:

$$D_1 = \tfrac{1}{2}(50) + \tfrac{1}{2}(40) = 25 + 20 = 45. \tag{11.2}$$

The old debt now has the same payoff, so $D_0$ also falls from 50 to 45.

**Answer (a).** $D_1 = \$45$.

**Step 3 — (b) Share price at the announcement.** Before the buyback the firm holds the machine plus the \$45 cash raised and owes debt worth $2 \times 45 = 90$; the 100 shares are still outstanding:

$$E = (145) + (45) - (90) = 100, \qquad P = \frac{100}{100} = 1.00. \tag{11.3}$$

The firm then buys back $n_R = 45/1.00 = 45$ shares; 55 remain. Equity pays $(210-100,\ 0) = (110,\ 0)$, worth $\frac{1}{2}(110) = 55 = 145 - 90$, i.e. still \$1.00 per share.

**Answer (b).** The share price jumps from \$0.95 to \$1.00.

**Assumption.** "A shareholder with 20 percent of the shares" holds 20 of the 100 shares before the announcement; "undoing the leverage change" means rebuilding the pre-announcement payoff exposure with personal trades.

**Step 4 — (c) Undoing the leverage.** Before, she held 20% of an equity paying $(160, 30)$, i.e. $(32, 6)$, worth $\frac{1}{2}(32) + \frac{1}{2}(6) = 16 + 3 = 19$. After the announcement her 20 shares are worth 20. One way to undo the change: keep $0.2 \times 55 = 11$ shares (paying $11 \times (2, 0) = (22, 0)$), sell $20 - 11 = 9$ shares at \$1.00, and with the \$9 buy 20% of the new debt, $0.2 \times 45 = 9$, which pays $0.2 \times (50, 40) = (10, 8)$. The position pays

$$(22 + 10,\ 0 + 8) = (32,\ 8), \qquad \text{worth } \tfrac{1}{2}(32) + \tfrac{1}{2}(8) = 20. \tag{11.4}$$

This beats the old payoff $(32, 6)$ state by state; the gain $20 - 19 = 1$ is 20% of the \$5 lost by the old creditors ($0.2 \times 5 = 1$).

To rebuild $(32, 6)$ exactly, hold $n$ shares (each paying $(2, 0)$) and a fraction $\beta$ of all the debt (paying $\beta(100, 80)$). The bad state gives $80\beta = 6$, so $\beta = 0.075$; the good state gives $2n + (0.075)(100) = 32$, so $2n = 24.5$ and $n = 12.25$. The cost is $12.25(1.00) + 0.075(90) = 12.25 + 6.75 = 19.00$, and she keeps \$1 in cash from her \$20 of shares.

**Answer (c).** She can rebuild her old payoff $(32, 6)$ for \$19 while her shares are worth \$20: she is \$1 better off.

**Step 5 — (d) MM still holds.** Total firm value is $50 + 95 = 145$ before and $90 + 55 = 145$ after, as (11.1) requires: changing the capital structure did not change the value of the claims taken together, and investors can undo it (Step 4). The share price rose only because equal-seniority debt diluted the old creditors, a transfer of \$5 from them to the shareholders, not value creation.

**Answer (d).** Firm value is \$145 before and after; the \$1.00 price reflects a \$5 transfer from old creditors, not new value.

**Intuition.** MM is a statement about the size of the pie, not about how it is cut. Issuing new debt that ranks equally with old debt cuts the old creditors' slice; without protective covenants, shareholders pocket that slice.

**Common mistakes.**

- Pricing the new debt at its face \$50, ignoring that equal seniority makes all the debt risky.
- Dividing post-repurchase equity by 100 shares (or announcement equity by 55): the announcement price uses 100 shares and the cash still in the firm.
- Concluding that MM fails because the share price rose: the old creditors lost exactly what shareholders gained.

**Book answer.** None in the book.

**Ref:** Grinblatt & Titman Ch. 14, §14.1 "The Modigliani-Miller Theorem" p. 501, §14.2 "How an Individual Investor Can 'Undo' a Firm's Capital Structure Choice" p. 505, §14.3 "How Risky Debt Affects the Modigliani-Miller Theorem" pp. 506–508.

## 12. Leveraged recapitalisation with corporate taxes — Berk 16.18, p. 493

**Statement.**

**Source:** Berk, Ch. 16, Exercise 16.18 (starred), p. 493 (PDF 524) · Answer in book: no

Kurz Manufacturing is currently an all-equity firm with 20 million shares outstanding and a stock price of \$7.50 per share. Although investors currently expect Kurz to remain an all-equity firm, Kurz plans to

announce that it will borrow $50 million and use the funds to repurchase shares. Kurz will pay interest only on this debt, and it has no further plans to increase or decrease the amount of debt. Kurz is subject to a 40% corporate tax rate.

a. What is the market value of Kurz's existing assets before the announcement?

b. What is the market value of Kurz's assets (including any tax shields) just after the debt is issued, but before the shares are repurchased?

c. What is Kurz's share price just before the share repurchase? How many shares will Kurz repurchase?

d. What are Kurz's market value balance sheet and share price after the share repurchase?

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**What is asked.** (a) The market value of Kurz's assets before the announcement. (b) The market value of its assets, tax shield included, after the $50 million is borrowed but before any shares are bought back. (c) The share price just before the buyback, and how many shares the $50 million buys. (d) The market-value balance sheet and share price after the buyback.

**Approach.** Build *market-value balance sheets* at each date. Permanent interest-only debt creates an interest tax shield worth $\tau_c D$ (*MM I with taxes*), which is capitalised into the share price as soon as the plan is known. The buyback then happens at that fair price, so it moves no value between selling and remaining shareholders.

**Notation.**

- $N_0 = 20$ million shares at $P_0 = 7.50$; all equity before the announcement.
- $D = 50$ million of permanent interest-only debt; $\tau_c = 0.40$; $A_1$: assets after the issue; $P_1$: share price after the debt is issued; $n_R$: shares repurchased (millions).

**Step 1 — (a) Existing assets.** With no debt, assets equal equity:

$$N_0 P_0 = (20)(7.50) = 150 \text{ million.}$$

**Answer (a).** $150 million.

**Step 2 — (b) Assets after the issue, before the repurchase.** The firm holds the old assets, the $50 million cash, and the tax shield of permanent debt, $\tau_c D = (0.40)(50) = 20$:

$$A_1 = (150) + (50) + (20) = 220 \text{ million.} \tag{12.1}$$

**Answer (b).** $220 million (existing assets 150, cash 50, tax shield 20).

**Step 3 — (c) Price and shares repurchased.** Equity is assets minus debt, spread over the 20 million shares still outstanding:

$$E_1 = A_1 - D = (220) - (50) = 170 \text{ million,}$$

$$P_1 = \frac{E_1}{N_0} = \frac{170}{20} = 8.50, \tag{12.2}$$

$$n_R = \frac{D}{P_1} = \frac{(50)}{(8.50)} = 5.882353 \text{ million shares.} \tag{12.3}$$

**Answer (c).** $P_1 = \$8.50$; Kurz repurchases 5.88 million shares.

**Step 4 — (d) After the repurchase.** The cash is gone and equity falls by the $50 million paid out:

| Assets | ($m) | Liabilities and equity | ($m) |
|---|---|---|---|
| Existing assets | 150 | Debt | 50 |
| Tax shield | 20 | Equity | 120 |
| Total | 170 | Total | 170 |

With $20 - 5.882353 = 14.117647$ million shares left, the price is

$$P_2 = \frac{120}{14.117647} = 8.50,$$

unchanged by the repurchase itself.

**Answer (d).** Assets $150 + 20 = 170$ = debt 50 + equity 120 ($m); share price $8.50.

**Intuition.** The whole tax benefit, $20 million or $1 per share, is capitalised into the price at the announcement. Shareholders who sell in the buyback and those who stay share it equally, since the repurchase is at the fair price $8.50.

**Common mistakes.**

- Repurchasing at $7.50 ($50/7.50 = 6.6667$ million shares): once the plan is known the price already includes the tax shield.
- Leaving the $50 million cash out of (b) (answering 170).
- Forgetting that the buyback itself does not change the price: only the announcement does.

**Book answer.** None in the book.

**Ref:** Berk Ch. 16, §16.3 "Debt and Taxes", pp. 473–479 (recapitalising to capture the tax shield).

## 13. Derive the MM adjusted cost of capital — Ross 16.19, p. 578

**Statement.**

**Source:** Ross, Cap. 16, Exercise 16.19 (DESAFIO), p. 578 (PDF 602) · Answer in book: no (selected answers in the web-only Apêndice C)

Em um mundo com impostos apenas sobre lucros da pessoa jurídica, mostre que o CMPC pode ser escrito como CMPC $= R_U \times [1 - T_C(D/V)]$.

**Context needed to solve it** *From Ross Cap. 16, "Impostos e Proposição I de M&M" and "Impostos, o CMPC e a Proposição II", p. 553 (PDF 577):*

Suponhamos que o custo de capital da empresa U seja de 10%. Chamaremos isso de **custo não alavancado de capital** e usaremos o símbolo $R_U$ para representá-lo. Podemos imaginar o $R_U$ como o custo de capital que uma empresa teria se não tivesse dívida. [...] O valor da empresa não alavancada ($V_U$) é simplesmente:

$$V_U = \frac{\text{Lajir} \times (1 - T_C)}{R_U}$$

O valor da empresa alavancada ($V_L$) é:

$$V_L = V_U + T_C \times D$$

[...] No Capítulo 14, vimos que, após considerarmos o efeito dos impostos, o CMPC é:

$$\text{CMPC} = (E/V) \times R_E + (D/V) \times R_D \times (1 - T_C)$$

Para calcular esse CMPC, precisamos saber qual é o custo do capital próprio. A Proposição II de M&M com impostos da pessoa jurídica afirma que o custo do capital próprio é:

$$R_E = R_U + (R_U - R_D) \times (D/E) \times (1 - T_C) \qquad [16.4]$$

*From Ross Cap. 16, Tabela 16.6 "Resumo de Modigliani e Miller", part II, p. 555 (PDF 579):*

II. O caso com impostos sobre lucros da pessoa jurídica

A. Proposição I com impostos: o valor da empresa alavancada ($V_L$) é igual ao valor da empresa não alavancada ($V_U$) mais o valor presente do benefício fiscal dos juros:

$$V_L = V_U + T_C \times D$$

onde $T_C$ é a alíquota de impostos da pessoa jurídica, e $D$ é o montante da dívida.

B. Proposição II com impostos: o custo do capital próprio ($R_E$) é:

$$R_E = R_U + (R_U - R_D) \times (D/E) \times (1 - T_C)$$

onde $R_U$ é o custo não alavancado de capital, isto é, o custo de capital da empresa caso não haja dívida.

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**What is asked.** Prove that, with corporate taxes only, Ross's after-tax WACC (CMPC) equals $R_U[1 - T_C(D/V)]$.

**Approach.** Combine three facts from Ross §16.4: the unlevered firm is a perpetuity $EBIT(1 - T_C)$ priced at $R_U$; *MM I with taxes*, $V_L = V_U + T_C D$; and the definition of the CMPC. Multiplying the CMPC by $V$ turns it into the total after-tax cash flow paid to investors net of the tax saving, which is the unlevered cash flow. A second route goes through *MM II with taxes*.

**Notation.**

- $EBIT$: constant perpetual operating income, fully paid out; $T_C \in [0, 1)$: corporate tax rate (no personal taxes).
- $D$: perpetual riskless debt with rate $R_D$; $E$: levered equity value; $V = E + D = V_L$: levered firm value; $V_U$: unlevered value.
- $R_U$: unlevered cost of capital; $R_E$: cost of levered equity; CMPC $= \frac{E}{V} R_E + \frac{D}{V} R_D(1 - T_C)$ (Ross's WACC). Assume $E > 0$, $V > 0$.

**Claim.** CMPC $= R_U\big[1 - T_C(D/V)\big]$.

**Step 1 — Valuation identities (Ross §16.4).** The unlevered firm's perpetual cash flow is $EBIT(1 - T_C)$, and $V_U = EBIT(1 - T_C)/R_U$; multiplying by $R_U$,

$$EBIT(1 - T_C) = R_U V_U. \tag{13.1}$$

The shield $T_C R_D D$ is a perpetuity as risky as the debt, worth $T_C R_D D / R_D = T_C D$ (cf. (3.2)), so MM I with taxes is

$$V = V_U + T_C D. \tag{13.2}$$

**Step 2 — Returns of the claims.** Shareholders receive the after-tax income left after interest, $(EBIT - R_D D)(1 - T_C)$, forever; debtholders receive $R_D D$, a perpetuity worth $D$ at rate $R_D$. Valuing the equity stream as a perpetuity, $E = (EBIT - R_D D)(1 - T_C)/R_E$, so

$$E\, R_E = (EBIT - R_D D)(1 - T_C). \tag{13.3}$$

**Step 3 — Multiply the CMPC by $V$.** From the definition,

$$\begin{aligned} V \cdot \text{CMPC} &= E\, R_E + D\, R_D(1 - T_C) \\ &= \big[(EBIT - R_D D)(1 - T_C)\big] + R_D D(1 - T_C) && \text{(substituting (13.3))} \\ &= EBIT(1 - T_C) - R_D D(1 - T_C) + R_D D(1 - T_C) && \text{(expanding)} \\ &= EBIT(1 - T_C) \\ &= \big[R_U V_U\big] && \text{(substituting (13.1))} \\ &= R_U\big[V - T_C D\big] && \text{(substituting (13.2)).} \end{aligned} \tag{13.4}$$

**Step 4 — Divide by $V > 0$.**

$$\text{CMPC} = R_U\, \frac{V - T_C D}{V} = R_U\Big[1 - T_C \frac{D}{V}\Big]. \qquad \blacksquare \tag{13.5}$$

**Answer (proof).** CMPC $= R_U\big[1 - T_C(D/V)\big]$, proved in (13.4)–(13.5).

**Step 5 — Same result through MM II.** First derive MM II with taxes. From (13.1) and (13.2), $EBIT(1-T_C) = R_U V_U = R_U(V - T_C D) = R_U(E + D - T_C D)$. Expand (13.3) and substitute:

$$\begin{aligned} E\,R_E &= EBIT(1-T_C) - R_D D(1-T_C) \\ &= R_U(E + D - T_C D) - R_D D(1-T_C) \qquad \text{(substituting)} \\ &= R_U E + R_U D(1-T_C) - R_D D(1-T_C) \\ &= R_U E + (R_U - R_D) D(1-T_C). \end{aligned}$$

Dividing by $E$ gives Ross's [16.4], $R_E = R_U + (R_U - R_D)\frac{D}{E}(1-T_C)$. Substitute it into the CMPC:

$$\begin{aligned} \text{CMPC} &= \tfrac{E}{V}\left[R_U + (R_U - R_D)\tfrac{D}{E}(1-T_C)\right] + \tfrac{D}{V} R_D (1-T_C) \\ &= \tfrac{E}{V} R_U + \tfrac{D}{V}(R_U - R_D)(1-T_C) + \tfrac{D}{V} R_D (1-T_C) \\ &= \tfrac{E}{V} R_U + \tfrac{D}{V} R_U (1-T_C) \\ &= R_U\left[\tfrac{E+D}{V} - T_C \tfrac{D}{V}\right] = R_U\left[1 - T_C \tfrac{D}{V}\right], \end{aligned}$$

since $E + D = V$.

**Step 6 — Numerical check with Exercise 3.** There $T_C D/V = 21{,}000/419{,}666.67 = 0.05003971$, so CMPC $= (0.15)(1 - 0.05003971) = 0.14249404$, and $(0.14249404)(419{,}666.67) = 59{,}800.00 = EBIT(1-T_C)$, as (13.4) requires. Directly: equity income is $(92{,}000 - 5{,}400)(0.65) = 56{,}290$, so $R_E = 56{,}290/359{,}666.67 = 0.156506$ (MM II gives the same: $0.15 + (0.06)(0.166821)(0.65) = 0.15 + 0.006506$), and with $E/V = 0.857029$, $D/V = 0.142971$,

$$\text{CMPC} = (0.857029)(0.156506) + (0.142971)(0.09)(0.65) = 0.134130 + 0.008364 = 0.142494.$$

**Intuition.** The levered firm's investors, taken together, receive the unlevered cash flow $EBIT(1-T_C)$ *plus* the tax saving; the CMPC nets the saving out of the debt cost, so it must discount the unlevered cash flow to the levered value. Since $V$ exceeds $V_U$ by $T_C D$, the rate falls below $R_U$ by exactly $R_U T_C D/V$. The formula relies on fixed perpetual debt; under a constant leverage ratio the shield is riskier and the formula changes.

**Common mistakes.**

- Using MM II without taxes, $R_E = R_U + (R_U - R_D)D/E$: then the CMPC does not reduce to the claim.
- Confusing $V$ with $V_U$ in $D/V$: the weights are levered market values.
- Applying the formula to a firm that keeps $D/V$ constant: it is derived for fixed perpetual debt.

**Book answer.** None in the local copy (selected answers are in the web-only Apêndice C).

**Ref:** Ross Cap. 16, §16.4, pp. 551–556 (Proposition I with taxes; taxes, WACC and Proposition II).

## 14. Miller's gain from leverage with personal taxes — Amaro de Matos 2.2.7 #1, p. 58

**Statement.**

**Source:** Amaro de Matos, Ch. 2, Exercise 2.2.7 #1, p. 58 (PDF 76) · Answer in book: yes, p. 255

Show that in the presence of personal income tax and corporate income tax, the gain from leverage to stockholders is given by the expression

$$G_L = \left[1 - \frac{(1-\tau_c)(1-\tau_s)}{1-\tau_b}\right] B$$

where $B$ denotes the market value of the debt in the leveraged firm.

**Context needed to solve it** *From Amaro de Matos §2.2 "Personal and Corporate Taxation" and §2.2.1 "Demand for Bonds", p. 49 (PDF 67):*

On the basis of Miller's argument, there is the underlying assumption that taxes are differentiated. Consider here that besides the tax on corporate profits under which interest payments are deductible, hereby denoted by $\tau_c$, there are also personal income taxes denoted by $\tau_b$ and $\tau_s$ that are applicable, respectively, to bonds and to shares. [...]

To study the demand for bonds, consider the existence of two types of bonds: fully tax-exempt bonds, with equilibrium rate of return $r_\circ$, and the other bonds, which are not tax-exempt and have a required rate of return $r_b$. The after-tax rate of return on these latter bonds is $(1-\tau_b)r_b$. There is demand for the nonexempt bonds provided that this after-tax rate of return is greater than $r_\circ$. This means that

$$r_b \geq \frac{r_\circ}{1-\tau_b}.$$

Consider a firm that raises equity at a cost of capital before taxes denoted by $\rho$. If this equity is within the same class of risk as the tax-exempt bonds, with rate of return $r_\circ$ as just described, one would require the same rate of return for the equity after paying the taxes due. Thus

$$r_\circ = (1-\tau_s)(1-\tau_c)\rho.$$

*From Amaro de Matos §2.2.3 "The Equilibrium", p. 52 (PDF 70):*

[...] To see why, notice that to raise a certain amount $K$ of capital, one can issue either debt or equity. In the first case, the stockholders pay back $r_b(1-\tau_c)K$, whereas in the second, they pay $\rho(1-\tau_c)K$. But since in equilibrium $r_b^* = \rho$, there is no relative advantage in one form of raising capital with respect to the other.

Alternatively, the relative advantage of leverage for stockholders, $G_L$, is written as

$$G_L = [\rho(1-\tau_c) - r_b^*(1-\tau_c)]K = \left[1 - \frac{(1-\tau_s)(1-\tau_c)}{1-\tau_b(B^*)}\right]\frac{r_\circ K}{(1-\tau_s)}.$$

---

**What is asked.** Prove that, with corporate tax $\tau_c$ and personal taxes $\tau_s$ on equity income and $\tau_b$ on interest, the value that leverage creates for stockholders is $G_L = [1-(1-\tau_c)(1-\tau_s)/(1-\tau_b)]B$.

**Approach.** Use the *homemade-leverage* replication of MM, now counting every tax on each income stream (Miller 1977). An investor can copy the after-tax income of the levered firm's equity by holding the unlevered firm's equity and borrowing personally; since interest she pays is deductible at $\tau_b$, she must borrow less than the firm's share of debt. No arbitrage then pins down $S_l$, and $G_L = V_l - V_u$ follows.

**Notation.**

- $X$: random operating income per period of two otherwise identical firms, $u$ (unlevered) and $l$ (levered), forever.
- $B$: market value of $l$'s perpetual riskless debt, which pays $rB$ per period; $r > 0$ is the bond rate.
- $\tau_c, \tau_s, \tau_b \in [0,1)$: corporate tax rate (interest deductible), personal tax rate on equity income, personal tax rate on interest income.
- $S_u = V_u$, $S_l$: equity values; $V_l = S_l + B$; $\alpha \in (0,1]$: fraction held; $L$: personal loan; $G_L = V_l - V_u$: gain from leverage.

**Assumption.** As in Amaro's §2.2 and Appendix answer, individuals borrow and lend at the bond rate $r$, and personal interest is taxed (lending) or deductible (borrowing) at $\tau_b$; there are no transaction costs and no arbitrage.

**Step 1 — Payoff of the levered equity.** The firm pays corporate tax on $X - rB$, and the holder pays $\tau_s$ on what she receives:

$$Y_l = \alpha(X - rB)(1-\tau_c)(1-\tau_s). \tag{14.1}$$

**Step 2 — Payoff of the unlevered equity.** The same taxes apply to the whole $X$:

$$Y_u = \alpha X(1-\tau_c)(1-\tau_s). \tag{14.2}$$

**Step 3 — Size of the personal loan.** A loan $L$ at $r$ costs $rL$ in interest, of which $\tau_b rL$ comes back as a tax deduction, so $rL(1-\tau_b)$ per period after tax. Choose $L$ so this equals the debt-service part of (14.1), $\alpha rB(1-\tau_c)(1-\tau_s)$:

$$\begin{aligned} rL(1-\tau_b) &= \alpha rB(1-\tau_c)(1-\tau_s) \\ L &= \alpha B\,\frac{(1-\tau_c)(1-\tau_s)}{1-\tau_b} \qquad \text{(dividing by } r(1-\tau_b)>0\text{).} \end{aligned} \tag{14.3}$$

**Step 4 — Replication.** Buying $\alpha$ of $u$ and borrowing $L$ pays, by (14.2) and (14.3),

$$\begin{aligned} Y_u - rL(1-\tau_b) &= \alpha X(1-\tau_c)(1-\tau_s) - \big[\alpha rB(1-\tau_c)(1-\tau_s)\big] \\ &= \alpha(X-rB)(1-\tau_c)(1-\tau_s) = Y_l, \end{aligned}$$

in every state and every period. Its cost today is $\alpha S_u - L$.

**Step 5 — Law of one price.** If $\alpha S_l > \alpha S_u - L$, a holder of $l$'s shares sells them, builds the replica and keeps the difference with unchanged future income. If $\alpha S_l < \alpha S_u - L$, a holder of $u$'s shares sells them, buys $\alpha$ of $l$'s shares and lends $L$ (after-tax interest $rL(1-\tau_b)$, restoring $Y_u$ by Step 4 read backwards), again pocketing the difference. No arbitrage therefore requires

$$\begin{aligned} \alpha S_l &= \alpha S_u - L \\ \alpha S_l &= \alpha S_u - \alpha B\,\frac{(1-\tau_c)(1-\tau_s)}{1-\tau_b} && \text{(substituting (14.3))} \\ S_l &= S_u - B\,\frac{(1-\tau_c)(1-\tau_s)}{1-\tau_b} && \text{(dividing by } \alpha\text{).} \end{aligned} \tag{14.4}$$

**Step 6 — Firm values and the gain.** Add $B$ to both sides of (14.4) and use $V_l = S_l + B$, $V_u = S_u$:

$$\begin{aligned} V_l &= V_u + B - B\,\frac{(1-\tau_c)(1-\tau_s)}{1-\tau_b}, \\ G_L = V_l - V_u &= \Big[1 - \frac{(1-\tau_c)(1-\tau_s)}{1-\tau_b}\Big]B. \qquad \blacksquare \end{aligned} \tag{14.5}$$

The gain goes to the stockholders who decide to lever: they collect $B$ at issue and keep shares worth $S_l$, a total of $V_l$ instead of $V_u$.

**Answer (proof).** $G_L = \big[1 - (1-\tau_c)(1-\tau_s)/(1-\tau_b)\big]B$, proved in (14.5).

**Step 7 — Special cases.** (i) $\tau_s = \tau_b$ (including no personal taxes): $G_L = \tau_c B$, the MM (1963) result, Amaro (2.7). (ii) Miller's equilibrium (2.11), $(1-\tau_c)(1-\tau_s) = 1-\tau_b$: $G_L = 0$, so no single firm gains from leverage. (iii) $(1-\tau_c)(1-\tau_s) > 1-\tau_b$: $G_L < 0$, and the firm should not borrow. Illustration: with $\tau_c = 0.34$, $\tau_s = 0.15$, $\tau_b = 0.30$, $(0.66)(0.85)/0.70 = 0.561/0.70 = 0.801429$, so $G_L = 0.198571\,B$, well below $\tau_c B = 0.34\,B$.

**Intuition.** Debt moves a dollar of income from the corporate-plus-equity tax channel, where it keeps $(1-\tau_c)(1-\tau_s)$, to the interest channel, where it keeps $(1-\tau_b)$. Leverage creates value only to the extent that the second channel is less taxed than the first.

**Common mistakes.**

- Borrowing $\alpha B$ (the firm's share of debt) instead of $L$ in (14.3): the personal deduction at $\tau_b$ changes the size of the loan that copies the firm's debt service.
- The sign of the loan: borrowing *reduces* the cost of the replica, so $\alpha S_l = \alpha S_u - L$.
- Concluding that $G_L = \tau_c B$ in general: that holds only when $\tau_s = \tau_b$.

**Book answer.** The final expression matches the book's answer (p. 255). The intermediate line printed there, "$\alpha S_l = \alpha S_u + \alpha B\frac{(1-\tau_c)(1-\tau_s)}{(1-\tau_b)}$", has a sign typo: it must be a minus, as in (14.4), since the investor *borrows* that amount and the book's own next line, $V_l = V_u + B[1 - \cdots]$, follows only with the minus.

**Ref:** Amaro de Matos Ch. 2, §2.2 "Personal and Corporate Taxation", pp. 48–52 (Miller equilibrium, $G_L$ on p. 52); Appendix C.2.2, p. 255.

# 15. MM I with perpetual debt, and when borrowing and lending rates differ — Amaro de Matos 2.1.5 #3–#4, p. 48

**Statement.**

**Source:** Amaro de Matos, Ch. 2, Exercises 2.1.5 #3 and #4, p. 48 (PDF 66) · Answer in book: yes, pp. 252–254

**[2.1.5 #3]** Modigliani and Miller (1958) developed their theory under some different assumptions. They assumed that debt had an infinite life and that each period the firm had to pay interest on debt equal to $rD$. Moreover, firms could deduct the interest on debt in computing taxable profits. The return to the stockholders is defined as the time average of an infinite stream of payoffs and given by $X$. How would you obtain proposition MM1 with and without taxes?

**[2.1.5 #4]** In the context of the previous exercise, suppose that an individual in the economy faces a borrowing interest rate $\bar{r}$ and a lending interest rate $\underline{r}$ such that $\bar{r} > r > \underline{r}$, where $r$ is the corporate interest rate. What happens to MM1?

**Context needed to solve it** *From Amaro de Matos §2.1.1 "The Irrelevancy Statement", pp. 40–42 (PDF 58–60) — MM1 without taxes:*

Assume that a firm starts at time $t = 0$ and has payoffs $\Gamma_j(t)$ in state $j$ at any point in time $t$, with $j = 1, 2, \ldots k(t)$ denoting the possible states of nature at time $t$.

In the setting to be considered here, it is assumed that markets do not allow for arbitrage opportunities and also that markets are perfect, in the sense that there are no frictions of any type, taxes included. Also, there are two ways to finance the firm: either through equity only or through equity together with debt. When debt is issued, the firm is said to be *levered*; otherwise it is *unlevered*.

Equityholders and debtholders have different claims on the final payoff of the firm. Debtholders, also known as *bondholders* or *creditors*, buy a debt contract issued by the firm giving them the claim to a face value $F$ at time $T$. If at that point in time the state of nature is such that $\Gamma_j(T)$ is less than $F$, the firm is said to go *bankrupt*, in which case the creditholder will receive $\Gamma_j(T)$, if positive. Let $V_j(T)$ denote the positive part of $\Gamma_j(T)$, or $V_j(T) \equiv [\Gamma_j(T)]^+$. [...] The payoff to a debt instrument in the $j$-th state of nature is thus

$$B_j(T) = \min\{F, V_j(T)\}.$$

If $K$ is the number of possible states of nature at time $T$, then, under the absence of arbitrage opportunities, there exists a strictly positive vector $\psi \in R^K$ such that the present value of the debt is

$$B = \sum_{j=1}^{K} \psi_j B_j(T).$$

Notice that, if there is no state leading to bankruptcy, $B_j(T) = F$ and the preceding expression implies that

$$B = F\sum_{j=1}^{K} \psi_j \Rightarrow B = \frac{F}{R}. \qquad (2.1)$$

[...] In other words, the payoff to equityholders in state $j$ is

$$S_j(T) = [\Gamma_j(T) - \min\{F, V_j(T)\}]^+ = V_j(T) - \min\{F, V_j(T)\} = \max[0, V_j(T) - F], \qquad (2.2)$$

implying that

$$V_j(T) = S_j(T) + B_j(T).$$

The present value of the equity is thus

$$S = \sum_{j=1}^{K} \psi_j S_j(T).$$

In this setting, it is clear that the value of the unlevered firm $V_u$ is equal to the value $V_l$ of a levered firm since

$$V_l = S + B = \sum_{j=1}^{K} \psi_j [B_j(T) + S_j(T)] = \sum_{j=1}^{K} \psi_j V_j(T) = V_u$$

This result is known as the first proposition of Modigliani and Miller and may be stated as

**Proposition 2.1.1 (MM1).** *The market value of any firm is independent of its capital structure*

$$V_l = V_u. \qquad (2.3)$$

*From Amaro de Matos §2.1.3 "The MM Propositions with Taxes", pp. 44–45 (PDF 62–63) — MM1 with taxes:*

[...] In this section it is assumed that firms pay taxes at a rate $\tau \in [0, 1]$.

Consider the case of a levered firm. In that case, let $T$ be the point in time where both taxes and the creditholders are paid. In case of bankruptcy, no taxes are paid; otherwise, the firm pays $\tau[V_j(T) - F]$. In other words, in any case the taxes are calculated in state $j$ as

$$\theta_j^\tau(T) = \tau[V_j(T) - F]^+.$$

[...] The debtholders will receive in state $j$

$$B_j(T) = \min\{F, [V_j(T) - \theta_j^\tau(T)]^+\} = \min\{F, V_j(T)\},$$

exactly the same payoff as without taxes. The equityholders are left in state $j$ with

$$S_j^\tau(T) = V_j(T) - B_j(T) - \theta_j^\tau(T). \qquad (2.5)$$

[...] Given the payoffs in the presence of taxes, the value of the unlevered firm is

$$V_u^\tau = (1 - \tau) \sum_{j=1}^{K} \psi_j V_j(T).$$

In the case of the levered firm, the firm generates for equity and debtholders the value

$$V_l^\tau = \sum_{j=1}^{K} \psi_j [S_j^\tau(T) + B_j(T)].$$

[...] In other words,

$$V_l^\tau = (1 - \tau) \sum_{j=1}^{K} \psi_j V_j(T) + \tau \sum_{j=1}^{K} \psi_j B_j(T) = V_u^\tau + \tau B.$$

This result is known as the proposition of Modigliani and Miller under taxes and is summarized as

**Proposition 2.1.3.** *Under the absence of arbitrage opportunities and in the presence of taxes, the value of the firm increases linearly with the level of debt as*

$$V_l^\tau = V_u^\tau + \tau B. \qquad (2.7)$$

*From Amaro de Matos §1.2.5 "Beta Values", p. 26 (PDF 44) — the symbol R in (2.1):*

[...] The return on the preceding portfolio will be denoted by $R$ and is given by $R = 1 + r$. Such a portfolio is risk-free by construction and, as one would expect, for each unit invested at time 0 it gives back $(1 + r)$ units at $t = 1$.

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**What is asked.** #3: Prove MM I in the original 1958 setting (perpetual debt paying $rD$ each period, income measured as a perpetual average $X$), both without taxes ($V_l = V_u$) and with deductible

interest ($V_l = V_u + \tau B$). #4: What becomes of MM I if individuals borrow at $\bar{r}$ and lend at $\underline{r}$, with $\bar{r} > r > \underline{r}$, while firms borrow at $r$.

**Approach.** Both parts use the *homemade-leverage* arbitrage. For each firm's equity, build a portfolio from the other firm's securities plus personal borrowing or lending that pays the same income in every state; no arbitrage then forces the two to cost the same, which gives two inequalities whose combination is MM I. In #4 personal borrowing is dearer and lending cheaper than corporate debt, so the replicas cost more and the two inequalities no longer meet: they leave a band.

**Notation.**

- $X$: random operating income per period (the time average of the stream), identical for firms $u$ (unlevered) and $l$ (levered); any distribution.
- $B$: market value of $l$'s perpetual riskless debt, paying $rB$ per period; $r$: corporate interest rate.
- $\tau \in [0, 1)$: corporate tax rate, interest deductible ($\tau = 0$ is the no-tax case); no personal taxes.
- $V_u = S_u$, $V_l = S_l + B$; $\alpha \in (0, 1]$: fraction of a firm held by an investor.
- #4 only: $\bar{r}$ and $\underline{r}$: individual borrowing and lending rates, $\bar{r} > r > \underline{r} > 0$.

**Assumption.** In #3 individuals borrow and lend at $r$, as firms do; there are no transaction costs. In #4, as in the book's answer, $\tau = 0$ and individuals can lend only at $\underline{r}$ (they cannot buy $l$'s bonds at yield $r$).

## #3 — MM I with and without taxes

**Step 1 — Payoffs.** Shareholders of $u$ pay tax on all of $X$; those of $l$ pay tax on $X$ net of interest. Each period they receive

$$Y_u = X(1-\tau), \qquad Y_l = (X - rB)(1-\tau). \tag{15.1}$$

**Step 2 — Replicating $l$'s equity.** Buy $\alpha$ of $u$ and borrow $\alpha(1-\tau)B$ at $r$. By (15.1) the income is

$$\alpha X(1-\tau) - r\,\alpha(1-\tau)B = \alpha(X - rB)(1-\tau) = \alpha Y_l,$$

at a cost of $\alpha S_u - \alpha(1-\tau)B$ today.

**Step 3 — Upper bound.** If $S_l > S_u - (1-\tau)B$, a holder of $\alpha$ of $l$'s shares sells them for $\alpha S_l$, builds the replica of Step 2 and keeps, with identical future income,

$$\begin{aligned} \alpha\big[S_l - S_u + (1-\tau)B\big] &= \alpha\big[(V_l - B) - V_u + (1-\tau)B\big] \qquad (\text{using } S_l = V_l - B,\ S_u = V_u) \\ &= \alpha\big[V_l - V_u - B + B - \tau B\big] \\ &= \alpha\big[V_l - V_u - \tau B\big] > 0, \end{aligned}$$

an arbitrage. Hence $V_l \le V_u + \tau B$.

**Step 4 — Replicating $u$'s equity.** Buy $\alpha$ of $l$'s shares and $\alpha(1-\tau)$ of its debt (or lend $\alpha(1-\tau)B$ at $r$):

$$\alpha(X - rB)(1-\tau) + \alpha(1-\tau)rB = \alpha X(1-\tau) = \alpha Y_u,$$

at a cost of $\alpha\big[S_l + (1-\tau)B\big]$.

**Step 5 — Lower bound.** If $S_u > S_l + (1-\tau)B$, then substituting $S_u = V_u$, $S_l = V_l - B$ gives $V_u > V_l - B + B - \tau B$, i.e. $V_l < V_u + \tau B$. A holder of $u$'s shares then sells them, buys the replica of Step 4 and keeps $\alpha[V_u + \tau B - V_l] > 0$. Hence $V_l \ge V_u + \tau B$.

**Step 6 — MM I.** Steps 3 and 5 give

$$V_l = V_u + \tau B \quad (\text{with taxes}), \qquad V_l = V_u \quad (\tau = 0). \qquad \blacksquare \tag{15.2}$$

**Answer (#3).** $V_l = V_u + \tau B$ with deductible interest; $V_l = V_u$ without taxes.

**Book answer.** Matches the book's answer (pp. 252–253), $V_u + \tau B = V_l$. The book's switching portfolio invests all the proceeds in the other firm instead of replicating exactly, which leads to the same two inequalities.

## #4 — Different borrowing and lending rates ($\tau = 0$)

**Step 7 — Replicating $l$'s equity with costly borrowing.** The replica must still hold $\alpha$ of $u$ (the only other claim on $X$), so its debt service has to equal $\alpha r B$ at the rate $\bar{r}$:

$$\bar{r}\,\bar{L} = \alpha r B \quad \Longrightarrow \quad \bar{L} = \alpha B\,\frac{r}{\bar{r}}, \tag{15.3}$$

which gives income $\alpha X - \bar{r}\bar{L} = \alpha X - \alpha r B = \alpha Y_l$ at a cost of $\alpha S_u - \bar{L} = \alpha\big(S_u - B\,r/\bar{r}\big)$. (Borrowing the full $\alpha B$ would cost $\alpha \bar{r} B > \alpha r B$ per period: not a replica.)

**Step 8 — Upper bound.** No arbitrage against an overvalued $l$ requires $\alpha S_l \le \alpha\big(S_u - B\,r/\bar{r}\big)$, i.e. $S_l \le S_u - B\,r/\bar{r}$. Add $B$ to both sides and use $V_l = S_l + B$, $V_u = S_u$:

$$V_l \le V_u + B - B\,\frac{r}{\bar{r}} = V_u + B\Big(1 - \frac{r}{\bar{r}}\Big) = V_u + B\,\frac{\bar{r} - r}{\bar{r}}. \tag{15.4}$$

**Step 9 — Replicating $u$'s equity with poor lending.** Buy $\alpha$ of $l$'s shares and lend $\underline{L}$ at $\underline{r}$ with

$$\underline{r}\,\underline{L} = \alpha r B \quad \Longrightarrow \quad \underline{L} = \alpha B\,\frac{r}{\underline{r}} > \alpha B, \tag{15.5}$$

giving income $\alpha(X - rB) + \alpha r B = \alpha Y_u$ at a cost of $\alpha\big(S_l + B\,r/\underline{r}\big)$.

**Step 10 — Lower bound.** No arbitrage against an overvalued $u$ requires $S_u \le S_l + B\,r/\underline{r}$. Substitute $S_u = V_u$ and $S_l = V_l - B$:

$$\begin{aligned} V_u &\le V_l - B + B\,\frac{r}{\underline{r}} \\ V_l &\ge V_u + B - B\,\frac{r}{\underline{r}} = V_u - B\Big(\frac{r}{\underline{r}} - 1\Big) = V_u - B\,\frac{r - \underline{r}}{\underline{r}}. \end{aligned} \tag{15.6}$$

**Step 11 — What happens to MM1.** Combining (15.4) and (15.6),

$$V_u - B\,\frac{r - \underline{r}}{\underline{r}} \;\le\; V_l \;\le\; V_u + B\,\frac{\bar{r} - r}{\bar{r}}. \qquad \blacksquare \tag{15.7}$$

MM1 no longer pins down $V_l$: arbitrage only confines it to a band around $V_u$ whose width grows with the debt $B$ and with the spreads. With $\bar{r} = r + \epsilon$, $\underline{r} = r - \epsilon$ the band is $V_u - \epsilon B/(r - \epsilon) \le V_l \le V_u + \epsilon B/(r + \epsilon)$, and it collapses to $V_l = V_u$ as $\epsilon \to 0$. Leverage can now raise firm value (up to the upper bound): the firm borrows more cheaply than individuals, so corporate leverage is a service investors cannot fully copy. If individuals could buy $l$'s bonds at yield $r$, Step 4 would apply and the lower bound would tighten to $V_l \ge V_u$.

**Answer (#4).** MM1 fails as an equality: $V_l$ is only confined to the band (15.7) around $V_u$, which shrinks to $V_l = V_u$ as the spreads vanish.

**Intuition.** MM1 rests on investors being able to copy corporate leverage on the same terms. When personal borrowing is dearer and lending cheaper than corporate debt, homemade leverage is an imperfect substitute: arbitrage still limits mispricing, but only to within the cost of the personal-borrowing wedge.

**Common mistakes.**

- In #3 with taxes, borrowing $\alpha B$ instead of $\alpha(1 - \tau)B$: the firm's interest is tax-deductible, so its equity bears only $(1 - \tau)rB$ of debt service.

- In #4, borrowing the full $\alpha B$ at $\bar{r}$ and calling it a replica: the debt service then differs, so the incomes do not match state by state.
- Concluding that MM1 "fails completely": arbitrage still bounds $V_l$, only less tightly.

**Book answer.** Differs from the book's printed band (p. 254), $V_u(1-\epsilon B_l/X) \le V_l \le V_u(1+\epsilon B_l/X)$. The qualitative conclusion is the same (a band around $V_u$, proportional to $\epsilon B$, vanishing at $\epsilon = 0$), but the bounds differ. The book's portfolios keep the investor's capital fixed (all proceeds go into the other firm, and the full principal $B_l$ is borrowed or lent), so the two incomes differ by a multiple of the random $X$ minus a fixed amount. Its inequality therefore holds only for a given value of $X$ (or in expectation), not state by state. The portfolios above match incomes exactly, so (15.7) is a true no-arbitrage band for any distribution of $X$. The two coincide only if $X/V_u$ happens to equal $\bar{r}$ (upper bound) or $\underline{r}$ (lower bound).

**Ref:** Amaro de Matos Ch. 2, §2.1.1–2.1.3, pp. 40–46 (MM1, MM2, MM with taxes); Appendix C.2.1, pp. 252–254.

## 16. Bank capital without taxes: does more equity lower ROE and raise funding costs? — Original exercise

**Statement.**

**Origin:** original exercise on the bank capital debate (Statements 1 and 2) and MM I/II without taxes, built on Prof. Ricca's Topic 2B slides 46–47 (Aula/Topic2B.pdf) and Topic 2A slides 89 and 97–102 (Aula/Topic2A.pdf) · Answer: in the solutions file

A bank holds loans with a market value of $A = 100$ ($ billion) at $t = 0$. One year later the loans pay $A(1+\tilde{r}_A)$, where the return on assets $\tilde{r}_A$ is 9% in a good state and 3% in a bad state, each with probability $\frac{1}{2}$. The loans are fairly priced: their required (expected) return is $r_A = 6\%$. The bank funds the loans with deposits $D$ and equity $E$, $D + E = A$. Deposits pay the risk-free rate $r_D = r_f = 4\%$ and are repaid at $t = 1$ before shareholders receive anything. There are no taxes, no bankruptcy costs and no other frictions, and investors can borrow and lend at $r_f$. The equity ratio is $e = E/A$, and the bank's return on equity is

$$ROE = \frac{(\text{payoff to shareholders at } t=1) - E}{E}.$$

The regulator considers raising the minimum equity ratio from $e = 4\%$ to $e = 20\%$; the bank keeps the same loans.

a. For $e = 4\%$ and $e = 20\%$, find $E$, $D$ and the amount owed to depositors at $t = 1$, and show that deposits are repaid in full in both states, so they are indeed risk-free.

b. For each equity ratio, compute the payoff to shareholders and the ROE in each state, the expected ROE $\mathbb{E}[ROE]$, and the standard deviation $\sigma(ROE)$. Compute also the ratio $(\mathbb{E}[ROE] - r_f)/\sigma(ROE)$ at each equity ratio.

c. Use MM II to compute the required return on equity $r_E$ at each equity ratio, compare it with the expected ROE of part (b), and compute the weighted average cost of capital $r_{WACC} = e\,r_E + (1-e)\,r_D$ at each ratio. Does the higher requirement change the value of the bank?

d. A shareholder who liked the ROE of the $e = 4\%$ bank holds shares of the bank after it moves to $e = 20\%$. Per $1 of her own money, how much of the 20% bank's equity must she buy and how much must she borrow at $r_f$ to receive exactly the ROE of the $e = 4\%$ bank in each state? Verify state by state.

e. Statement 1 (slide 47): "Increased equity requirements will hurt bank shareholders since it would lower the banks' return on equity (ROE)." True or false? Justify with your answers to (b)–(d).

f. Statement 2 (slide 47): "Increased equity requirements increase the funding costs for banks because they must use more equity, which has a higher required return." A bank executive supports it with this computation: "At $e = 4\%$ our equity costs $r_E^{4\%}$ (the value you found in (c)) and our deposits 4%; at 20% equity our funding cost becomes $0.20\,r_E^{4\%} + 0.80 \times 4\%$, against 6% today." Compute the executive's number, find the error, and say whether Statement 2 is true or false.

**Context needed to solve it** *From Prof. Ricca's Topic 2B slide 46 (Aula/Topic2B.pdf):*

Admati et al. (2013): no! Bank equity is not socially expensive. They use many of the arguments discussed in this lecture. They call for larger capital requirements, that is, less leverage in the banking industry.

*From Prof. Ricca's Topic 2A slides 98–99 and 101–102 (Aula/Topic2A.pdf) — MM I and MM II:*

MM I implies that the market value of the firm, $V$, or equivalently, the market value of the assets of the firm, $A$, is independent of the choices of $D$ and $E$. As the expected return on assets is defined by $A = \mathbb{E}[\tilde{C}_A]/(1 + r_A)$ and $A$ and $\tilde{C}_A$, the cash flows generated by the assets, are independent of $D$ and $E$, $r_A$ is independent of $D$ and $E$. Recall the identity

$$r_A = \frac{E}{E+D}\, r_E + \frac{D}{E+D}\, r_D.$$

Rearranging, we have that

$$r_E = r_A + \frac{D}{E}\,(r_A - r_D).$$

As $r_A$ is fixed (independent of $E$ and $D$) and $r_A > r_D$, $r_E$ increases with $D/E$, the market debt to equity ratio. For small values of $D$, that is, when debt is risk-free, $r_D$ is fixed and equal to $r_f$, and $r_E$ is linear in $D/E$. [...] We refer to $r_A$ as the weighted average cost of capital, $r_{WACC}$. MM I implies that $r_{WACC}$ does not depend on how the firm finances its assets (on the choices of $D$ and $E$). [...] Conservation of value principle: With perfect capital markets, financial transactions neither add nor destroy value, but instead represent a repackaging of risk (and therefore return).

*From Prof. Ricca's Topic 2A slide 89 (Aula/Topic2A.pdf) — homemade leverage:*

The arbitrageur can "lever up" firm 2 by borrowing on individual accounts (homemade leverage). Absence of arbitrage implies that $V_1 = V_2$.

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*Original exercise (list, Exercise 16), built on Topic 2B slides 46–47 and Topic 2A slides 89 and 97–102. Amounts in $ billion.*

**What is asked.** (a) The balance sheet at the two equity ratios and a check that deposits are really safe. (b) What shareholders earn per dollar invested (ROE) in each state, its mean and its risk, and the reward per unit of risk. (c) The return shareholders *require* at each ratio (MM II), whether it matches what they expect to earn, and whether the overall cost of funding (WACC) or the bank's value moves. (d) How a shareholder can rebuild the high-leverage ROE herself by borrowing. (e)–(f) Whether Statements 1 and 2 of slide 47 survive these numbers.

**Approach.** The bank is a firm in a perfect market, so MM applies unchanged: the loans' cash flows and their required return $r_A$ do not depend on how they are funded (*MM I*). Leverage only redistributes the same risk: with less equity, each dollar of equity carries more of the asset risk and therefore must earn more (*MM II*). ROE compares returns on claims of different risk, so a lower ROE at higher capital is not a loss. *Homemade leverage* shows the old ROE remains available to any investor who wants it.

**Notation.**

- $A = 100$: market value of the loans at $t = 0$; $\tilde{r}_A \in \{9\%, 3\%\}$: return on assets (good, bad state), probability $\frac{1}{2}$ each; $r_A = 6\%$: required return on assets; $\sigma_A$: standard deviation of $\tilde{r}_A$.
- $e = E/A$: equity ratio; $E$, $D = A - E$: equity and deposits; $r_D = r_f = 4\%$: deposit (risk-free) rate.
- $\Pi_s$: payoff to shareholders at $t = 1$ in state $s \in \{g, b\}$; $ROE_s = \Pi_s/E - 1$; $r_E$: required return on equity; $r_{WACC}$: weighted average cost of capital.

**Step 1 — (a) Balance sheets and the repayment owed.** The equity ratio fixes $E = eA$, deposits are the rest, and depositors are owed principal plus interest:

$$\begin{aligned} e = 4\%: &\quad E = (0.04)(100) = 4, \quad D = 100 - 4 = 96, \quad D(1 + r_f) = (96)(1.04) = 99.84, \quad (16.1)\\ e = 20\%: &\quad E = (0.20)(100) = 20, \quad D = 100 - 20 = 80, \quad D(1 + r_f) = (80)(1.04) = 83.2. \end{aligned}$$

**Step 2 — (a) Deposits are safe.** The worst case for depositors is the bad state, where the loans pay

$$A(1+0.03) = (100)(1.03) = 103. \tag{16.2}$$

Since $103 > 99.84$ and $103 > 83.2$, depositors are paid in full in both states at both ratios, so the deposits are risk-free and $r_D = r_f$ is the right rate. In the good state the loans pay $(100)(1.09) = 109$.

**Answer (a).** $e = 4\%$: $E = 4$, $D = 96$, owed 99.84; $e = 20\%$: $E = 20$, $D = 80$, owed 83.2; both below the bad-state asset payoff of 103, so deposits are risk-free.

**Step 3 — (b) Shareholders' payoffs.** Shareholders receive what is left after depositors:

$$\begin{aligned} e = 4\%: &\quad \Pi_g = 109 - 99.84 = 9.16, \qquad \Pi_b = 103 - 99.84 = 3.16, \\ e = 20\%: &\quad \Pi_g = 109 - 83.2 = 25.8, \qquad \Pi_b = 103 - 83.2 = 19.8. \end{aligned}$$

**Step 4 — (b) ROE in each state.** Divide each payoff by the equity invested and subtract 1:

$$\begin{aligned} e = 4\%: &\quad ROE_g = \frac{9.16}{4} - 1 = 2.29 - 1 = 129\%, \qquad ROE_b = \frac{3.16}{4} - 1 = 0.79 - 1 = -21\%, \\ e = 20\%: &\quad ROE_g = \frac{25.8}{20} - 1 = 1.29 - 1 = 29\%, \qquad ROE_b = \frac{19.8}{20} - 1 = 0.99 - 1 = -1\%. \end{aligned}$$

The same numbers follow from $ROE_s = \tilde{r}_{A,s} + \frac{D}{E}(\tilde{r}_{A,s} - r_f)$: at $e = 4\%$, $D/E = 96/4 = 24$ and $ROE_g = 9\% + 24(9\% - 4\%) = 9\% + 120\% = 129\%$.

**Step 5 — (b) Mean, risk and reward per unit of risk.** With two equally likely states the mean is the average and the standard deviation is half the gap between the two outcomes:

$$\begin{aligned} e = 4\%: &\quad \mathbb{E}[ROE] = \tfrac{1}{2}(129\%) + \tfrac{1}{2}(-21\%) = 54\%, \\ &\quad \sigma(ROE) = \tfrac{1}{2}\,|129\% - (-21\%)| = \tfrac{1}{2}(150\%) = 75\%, \\ e = 20\%: &\quad \mathbb{E}[ROE] = \tfrac{1}{2}(29\%) + \tfrac{1}{2}(-1\%) = 14\%, \\ &\quad \sigma(ROE) = \tfrac{1}{2}\,|29\% - (-1\%)| = \tfrac{1}{2}(30\%) = 15\%. \end{aligned} \tag{16.3}$$

For comparison, $\sigma_A = \frac{1}{2}|9\% - 3\%| = 3\%$, and $\sigma(ROE) = (A/E)\,\sigma_A$: $(25)(3\%) = 75\%$ and $(5)(3\%) = 15\%$. The reward per unit of risk is

$$e = 4\%: \quad \frac{54\% - 4\%}{75\%} = \frac{50}{75} = 0.6667, \qquad e = 20\%: \quad \frac{14\% - 4\%}{15\%} = \frac{10}{15} = 0.6667. \tag{16.4}$$

**Answer (b).** $e = 4\%$: ROE 129% / −21%, $\mathbb{E}[ROE] = 54\%$, $\sigma = 75\%$; $e = 20\%$: ROE 29% / −1%, $\mathbb{E}[ROE] = 14\%$, $\sigma = 15\%$. The ratio is 0.6667 at both: expected ROE falls exactly in proportion to risk.

**Step 6 — (c) Required return on equity (MM II).** With safe debt, $r_E = r_A + \frac{D}{E}(r_A - r_D)$:

$$\begin{aligned} e = 4\%: &\quad r_E = 6\% + \Big(\frac{96}{4}\Big)(6\% - 4\%) = 6\% + (24)(2\%) = 6\% + 48\% = 54\%, \\ e = 20\%: &\quad r_E = 6\% + \Big(\frac{80}{20}\Big)(6\% - 4\%) = 6\% + (4)(2\%) = 6\% + 8\% = 14\%. \end{aligned} \tag{16.5}$$

These equal the expected ROEs in (16.3): at both ratios shareholders expect to earn exactly what their risk requires, so their claim has zero NPV either way. Indeed the equity is worth $\mathbb{E}[\Pi]/(1 + r_E)$: $\frac{1}{2}(9.16 + 3.16)/1.54 = 6.16/1.54 = 4$ and $\frac{1}{2}(25.8 + 19.8)/1.14 = 22.8/1.14 = 20$, the amounts invested.

**Step 7 — (c) WACC and value.** Weight each required return by its share of funding:

$$\begin{aligned} e = 4\%: &\quad r_{WACC} = (0.04)(54\%) + (0.96)(4\%) = 2.16\% + 3.84\% = 6\%, \\ e = 20\%: &\quad r_{WACC} = (0.20)(14\%) + (0.80)(4\%) = 2.80\% + 3.20\% = 6\%. \end{aligned} \tag{16.6}$$

Both equal $r_A$. The bank's value is $E + D = 4 + 96 = 20 + 80 = 100 = A$ at both ratios.

**Answer (c).** $r_E = 54\%$ at 4% equity and 14% at 20% equity, equal to the expected ROEs; $r_{WACC} = 6\%$ at both; the bank's value stays 100.

**Step 8 — (d) Homemade leverage.** One dollar of the 4% bank's equity carries $A/E = 1/0.04 = 25$ dollars of loans; one dollar of the 20% bank's equity carries $1/0.20 = 5$. To rebuild 25 dollars of loan exposure per dollar of her own money, she buys $25/5 = 5$ dollars of the 20% bank's equity, financed with her \$1 plus $5 - 1 = 4$ dollars borrowed at $r_f$. Her return in each state:

$$\begin{aligned} \text{good:} \quad & \frac{(5)(1.29) - (4)(1.04)}{1} - 1 = 6.45 - 4.16 - 1 = 1.29 = 129\%, \\ \text{bad:} \quad & \frac{(5)(0.99) - (4)(1.04)}{1} - 1 = 4.95 - 4.16 - 1 = -0.21 = -21\%. \end{aligned} \tag{16.7}$$

**Answer (d).** Buy \$5 of the 20% bank's equity per \$1 of her own money, borrowing \$4 at 4%; she gets 129% and −21%, exactly the 4% bank's ROE.

**Step 9 — (e) Statement 1.** Expected ROE does fall, from 54% to 14%, but risk falls in the same proportion (16.4), the required return falls by the same amount (16.5), the value of the shareholders' claim equals what they paid at both ratios, and the high-ROE payoff is still available through (16.7).

**Answer (e).** False. A lower ROE is the price of lower risk, not a loss to shareholders.

**Step 10 — (f) Statement 2.** The executive's computation keeps $r_E$ at its 4%-equity level:

$$(0.20)(54\%) + (0.80)(4\%) = 10.8\% + 3.2\% = 14\%. \tag{16.8}$$

But $r_E$ is not a constant: by MM II it falls to 14% when equity rises to 20%, and the correct WACC (16.6) stays at 6%. Equity has a higher required return than debt at every ratio, yet replacing debt with equity does not raise the funding cost, because it makes all equity less risky.

**Answer (f).** The executive's number is 14%; the error is holding $r_E$ fixed at 54%. Statement 2 is false: the funding cost stays at 6%.

**Intuition.** The loans' risk is fixed by the loans, not by the liability side. Leverage concentrates that risk on a thin layer of equity, which then demands a high return; more equity spreads the same risk over more dollars, so each dollar demands less. ROE is a return on a risky claim, and comparing ROEs across leverage levels compares claims of different risk.

**Common mistakes.**

- Judging shareholder welfare by expected ROE instead of by the value of their claim (or ROE per unit of risk).
- Using a fixed $r_E$ when leverage changes, as in (16.8).
- Forgetting to check that deposits are safe; MM II in the linear form needs $r_D = r_f$.

**Ref:** Prof. Ricca, Topic 2B slides 46–47 (bank capital debate, Statements 1–2); Topic 2A slide 89 (homemade leverage) and slides 97–102 (MM II, $r_{WACC}$ independent of $D$ and $E$, conservation of value).

## 17. Bank capital with corporate taxes: the lost interest tax shield — Original exercise

**Statement.**

**Origin:** original exercise on the bank capital debate (Statement 3) and MM I with corporate taxes, built on Prof. Ricca's Topic 2B slides 7–18 and 47 (Aula/Topic2B.pdf) · Answer: in the solutions file

A bank's assets generate a perpetual expected pre-tax cash flow of $X = 7.5$ (\$ billion per year). The corporate tax rate is $\tau = 20\%$, so the all-equity (unlevered) bank would pay $\tau X$ in taxes every year and its owners would receive $(1 - \tau)X = 6$ per year. The unlevered cost of capital is $r_U = 6\%$, so the unlevered bank

is worth $V_U = (1-\tau)X/r_U = 100$. The bank's debt is perpetual, risk-free and kept at a constant amount $D$; it pays $r_D = 4\%$, and interest is deductible from taxable income. The bank always has enough income to use the deduction in full. There are no bankruptcy costs and no other frictions. The equity ratio is measured at market value, $e = E/V_L$ with $V_L = D + E$ the value of the levered bank. The regulator raises the minimum equity ratio from $e = 5\%$ to $e = 20\%$; the bank keeps the same assets.

a. For $e = 5\%$ and $e = 20\%$, find $V_L$, $D$, $E$ and the value of the interest tax shield $V_{ITS}$. (Use $D = (1-e)V_L$ together with MM I with taxes.)

b. The bank moves from $e = 5\%$ to $e = 20\%$ by issuing new shares at a fair price and using all the proceeds to retire debt at its market value. How much debt is retired? Compute the annual interest tax saving $\tau r_D D$ at each ratio, the value lost by the bank's existing shareholders, and that loss as a fraction of their equity before the change.

c. Compute the after-tax WACC at each ratio with the constant-debt formula, and check that discounting the unlevered after-tax cash flow $(1-\tau)X$ at it gives $V_L$. Derive the required return on equity $r_E$ at each ratio from the market-value balance sheet with $r_{ITS} = r_D$, and check that $\frac{E}{V_L} r_E + \frac{D}{V_L} r_D(1-\tau)$ equals the after-tax WACC.

d. The government collects $\tau(X - r_D D)$ per year from the bank. The part $\tau X$ is as risky as the assets (discount it at $r_U$); the part $\tau r_D D$ is risk-free (discount it at $r_D$). Compute the present value $G$ of the government's tax claim at each ratio, and the total $V_L + G$. Who gains what the existing shareholders lose in (b)?

e. Statement 3 (slide 47): "Increased equity requirements increase the funding costs for banks because they reduce the ability of banks to benefit from the tax shield associated with interest payments on debt." Is it true for the bank's shareholders? Is the lost tax shield a cost for society as a whole? Explain why this argument does not show that equity is socially expensive.

**Context needed to solve it** *From Prof. Ricca's Topic 2B slides 7–8 (Aula/Topic2B.pdf) — MM I with corporate taxes:*

Relative to an identical unlevered firm, the levered firm can pay out $\tau r_f D$ more to its investors in each period. Assuming perpetuity

$$V_{ITS} = \frac{\tau r_f D}{r_f} = \tau D.$$

(MM I with corporate taxes) The value of a levered firm equals that of an unlevered firm plus the present value of the interest tax shield

$$V_L = V_U + V_{ITS}.$$

If investors cannot get an equivalent tax break on homemade leverage, they are willing to pay a premium for levered firms. All the tax benefits from leverage accrue to the shareholders.

*From Prof. Ricca's Topic 2B slides 9–11 (Aula/Topic2B.pdf) — market-value balance sheet with taxes:*

Assets of the levered firm: $V_U$ (market value of the unlevered firm's assets, return $r_U$) and $V_{ITS}$ (return $r_{ITS}$); claims: debt $D$ (return $r_D$) and equity $E$ (return $r_E$); $V_L = V_U + V_{ITS} = D + E$. From the balance sheet, we have the identity

$$\frac{D}{D+E} r_D + \frac{E}{D+E} r_E = \frac{V_U}{V_U + V_{ITS}} r_U + \frac{V_{ITS}}{V_U + V_{ITS}} r_{ITS}.$$

$r_{ITS} = r_D$ if the firm maintains a constant market value debt amount in perpetuity. In this case, $V_{ITS} = \tau D r_D / r_D = \tau D$.

*From Prof. Ricca's Topic 2B slides 16 and 18 (Aula/Topic2B.pdf) — after-tax WACC:*

With tax-deductible interest, the effective return on debt is $r_D(1-\tau)$. The after-tax WACC is

$$r_{WACC} = \frac{E}{E+D} r_E + \frac{D}{E+D} r_D(1-\tau).$$

If we discount the unlevered free cash flows with the after-tax WACC, we obtain the correct value of the levered firm. Constant debt level:

$$r_{WACC} = \left(1 - \tau \frac{D}{D+E}\right) r_U.$$

*Original exercise (list, Exercise 17), built on Topic 2B slides 7–18 and 47. Amounts in $ billion, flows per year.*

**What is asked.** (a) The bank's market-value balance sheet at 5% and 20% equity when interest is tax-deductible. (b) What the move to 20% requires (debt retired) and what it costs the bank's current owners. (c) How the after-tax WACC and the required return on equity change, with two consistency checks. (d) What happens to the government's tax claim and to the total value of all claims. (e) Whether the lost tax shield (Statement 3) is a private cost, a social cost, or both.

**Approach.** With constant perpetual safe debt, *MM I with taxes* gives $V_L = V_U + \tau D$ (slides 7–8): the interest tax shield is the only thing leverage adds. Because the requirement is stated as a share of $V_L$, which itself depends on $D$, part (a) is a fixed point solved in one line. Part (d) counts the government as a third claimant: the pie $X$ is split among shareholders, debtholders and the government, and leverage only moves value between them.

**Notation.**

- $X = 7.5$: expected pre-tax cash flow per year (perpetual); $\tau = 20\%$: tax rate; $r_U = 6\%$: unlevered cost of capital; $V_U = (1-\tau)X/r_U = 100$.
- $D$: perpetual constant safe debt, $r_D = 4\%$; $E$: equity; $V_L = D + E$; $e = E/V_L$; $V_{ITS} = \tau D$.
- $r_E$: required return on equity; $r_{WACC}$: after-tax WACC; $G$: present value of the government's tax claim.

**Step 1 — (a) Solve for $V_L$.** Substitute $D = (1-e)V_L$ into $V_L = V_U + \tau D$ and collect $V_L$:

$$\begin{aligned} V_L &= V_U + \tau(1-e)V_L \\ V_L\big[1-\tau(1-e)\big] &= V_U \\ V_L &= \frac{V_U}{1-\tau(1-e)}. \end{aligned} \tag{17.1}$$

**Step 2 — (a) Balance sheet at $e = 5\%$.**

$$\begin{aligned} V_L &= \frac{100}{1-(0.20)(0.95)} = \frac{100}{1-0.19} = \frac{100}{0.81} = 123.4568, \\ D &= (0.95)(123.4568) = 117.2840, \qquad E = (0.05)(123.4568) = 6.1728, \\ V_{ITS} &= (0.20)(117.2840) = 23.4568 = V_L - V_U. \end{aligned}$$

**Step 3 — (a) Balance sheet at $e = 20\%$.**

$$\begin{aligned} V_L &= \frac{100}{1-(0.20)(0.80)} = \frac{100}{1-0.16} = \frac{100}{0.84} = 119.0476, \\ D &= (0.80)(119.0476) = 95.2381, \qquad E = (0.20)(119.0476) = 23.8095, \\ V_{ITS} &= (0.20)(95.2381) = 19.0476 = V_L - V_U. \end{aligned}$$

**Answer (a).** $e = 5\%$: $V_L = 123.457$, $D = 117.284$, $E = 6.173$, $V_{ITS} = 23.457$; $e = 20\%$: $V_L = 119.048$, $D = 95.238$, $E = 23.810$, $V_{ITS} = 19.048$.

**Step 4 — (b) Debt retired and the annual tax saving.** The debt retired equals the new equity raised:

$$\Delta D = 117.2840 - 95.2381 = 22.0459.$$

The annual interest tax saving $\tau r_D D$ is

$$e = 5\%: \ (0.20)(0.04)(117.2840) = 0.9383, \qquad e = 20\%: \ (0.20)(0.04)(95.2381) = 0.7619,$$

a fall of $0.9383 - 0.7619 = 0.1764$ per year.

**Step 5 — (b) Loss to existing shareholders.** New shareholders pay a fair price, so they pay exactly what their shares are worth and neither gain nor lose. The fall in $V_L$ therefore lands entirely on the existing shareholders:

$$\Delta V_L = 123.4568 - 119.0476 = 4.4092 = \tau\,\Delta D = (0.20)(22.0459), \tag{17.2}$$

which is also the perpetual annual loss discounted at $r_D$: $0.1764/0.04 = 4.41$. The existing shareholders keep $23.8095 - 22.0459 = 1.7637 = 6.1728 - 4.4092$, so the loss as a fraction of their equity is

$$\frac{4.4092}{6.1728} = 0.7143 = 71.43\%.$$

**Answer (b).** Retire 22.046 of debt; the annual tax saving falls from 0.938 to 0.762; existing shareholders lose 4.409, i.e. 71.43% of their pre-change equity.

**Step 6 — (c) After-tax WACC.** Constant-debt formula $r_{WACC} = \big(1-\tau\frac{D}{V_L}\big)r_U$, with $D/V_L = 1-e$:

$$\begin{aligned}
e = 5\%: \quad & r_{WACC} = \big(1-(0.20)(0.95)\big)(6\%) = (0.81)(6\%) = 4.86\%,\\
& \frac{(1-\tau)X}{r_{WACC}} = \frac{6}{0.0486} = 123.4568,\\
e = 20\%: \quad & r_{WACC} = \big(1-(0.20)(0.80)\big)(6\%) = (0.84)(6\%) = 5.04\%,\\
& \frac{(1-\tau)X}{r_{WACC}} = \frac{6}{0.0504} = 119.0476.
\end{aligned}$$

Both reproduce $V_L$ from Steps 2–3. The after-tax WACC rises by $5.04\% - 4.86\% = 0.18$ percentage points.

**Step 7 — (c) Required return on equity.** The balance-sheet identity (slide 10) multiplied by $V_L$, with $r_{ITS} = r_D$ and $V_{ITS} = \tau D$, is $E\,r_E + D\,r_D = V_U r_U + \tau D\,r_D$. Substitute $V_U = V_L - \tau D = E + (1-\tau)D$ and solve for $r_E$:

$$\begin{aligned}
E\,r_E &= \big[E + (1-\tau)D\big]r_U + \tau D\,r_D - D\,r_D\\
&= E\,r_U + (1-\tau)D\,r_U - (1-\tau)D\,r_D\\
r_E &= r_U + (1-\tau)\frac{D}{E}(r_U - r_D).
\end{aligned} \tag{17.3}$$

With $D/E = 0.95/0.05 = 19$ and $D/E = 0.80/0.20 = 4$:

$$\begin{aligned}
e = 5\%: \quad & r_E = 6\% + (0.80)(19)(2\%) = 6\% + 30.4\% = 36.4\%,\\
e = 20\%: \quad & r_E = 6\% + (0.80)(4)(2\%) = 6\% + 6.4\% = 12.4\%.
\end{aligned}$$

Check with the after-tax WACC formula (slide 16):

$$\begin{aligned}
e = 5\%: \quad & (0.05)(36.4\%) + (0.95)(4\%)(0.80) = 1.82\% + 3.04\% = 4.86\%,\\
e = 20\%: \quad & (0.20)(12.4\%) + (0.80)(4\%)(0.80) = 2.48\% + 2.56\% = 5.04\%.
\end{aligned}$$

**Answer (c).** After-tax WACC 4.86% → 5.04% (it gives back $V_L$); $r_E$ 36.4% → 12.4%; both checks hold.

**Step 8 — (d) The government's claim.** Discount $\tau X$ at $r_U$ and $\tau r_D D$ at $r_D$:

$$G = \frac{\tau X}{r_U} - \frac{\tau r_D D}{r_D} = \frac{(0.20)(7.5)}{0.06} - \tau D = 25 - \tau D, \tag{17.4}$$

$$\begin{aligned}
e = 5\%: \quad & G = 25 - 23.4568 = 1.5432, \qquad V_L + G = 123.4568 + 1.5432 = 125,\\
e = 20\%: \quad & G = 25 - 19.0476 = 5.9524, \qquad V_L + G = 119.0476 + 5.9524 = 125.
\end{aligned}$$

The total $125 = X/r_U = 7.5/0.06$ is the pre-tax value of the assets, independent of $D$. The government gains $5.9524 - 1.5432 = 4.4092$, exactly the shareholders' loss in (17.2).

**Answer (d).** $G = 1.543$ at 5% and 5.952 at 20%; $V_L + G = 125$ at both; the taxpayers gain the 4.409 the shareholders lose.

**Step 9 — (e) Statement 3.** For the bank's shareholders the statement is true: they lose the subsidy 4.409 and the after-tax WACC rises. For society it is a transfer, not a cost: the total value of the claims on the assets stays at 125, and the government's extra revenue can fund spending or lower taxes elsewhere. The lost tax shield shows that the tax code subsidises bank debt, not that equity uses up resources.

**Answer (e).** True privately, false socially: the lost tax shield is a transfer from bank shareholders to taxpayers, so it does not make equity socially expensive.

**Intuition.** Taxes add a third claimant to the bank. Debt lowers what the government takes, and that is the only reason it raises $V_L$. A capital requirement moves that slice back to the government. Whether banks should get the subsidy is a separate tax-policy question; it is no reason to keep leverage high.

**Common mistakes.**

- Treating $e$ as $E/V_U$ instead of $E/V_L$, or computing $D = (1-e)V_U$; the requirement is a fixed point in $V_L$ (17.1).
- Counting the lost tax shield as a loss to society; it reappears in $G$.
- Holding $r_E$ at 36.4% when equity rises (the Statement 2 mistake, again).

**Ref:** Prof. Ricca, Topic 2B slides 7–8 (MM I with taxes, $V_{ITS} = \tau D$), slides 9–11 (market-value balance sheet, $r_{ITS} = r_D$ for constant debt), slides 16–18 (after-tax WACC, constant-debt formula), slide 47 (Statement 3).

## 18. The failure externality and the socially optimal capital ratio — Original exercise

**Statement.**

**Origin:** original exercise on the bank capital debate (failure externality) and the static trade-off, built on Prof. Ricca's Topic 2B slides 44–47 (Aula/Topic2B.pdf) · Answer: in the solutions file

Measure everything per \$1 of bank assets per year. The bank's equity ratio is $e \in [0,1]$ and its deposits are $1-e$. Each year the assets suffer a loss rate $\ell \geq 0$ (loss as a fraction of assets) that is exponentially distributed with mean $\theta = 0.01$:

$$\Pr(\ell > x) = \exp(-x/\theta), \qquad x \geq 0.$$

The bank fails when the loss exceeds its equity, $\ell > e$. A failure imposes on the rest of the economy (the deposit insurer and taxpayers, and the disruption of credit to borrowers) a social loss of $\kappa = 0.50$ per \$1 of the bank's assets. The bank's shareholders bear none of it: they have limited liability, and deposits are fully insured at a flat premium, so the deposit rate does not depend on $e$. Deposits also give depositors liquidity services worth $\lambda = 0.005$ per \$1 of deposits per year; this is a genuine social benefit of deposit funding, and the bank captures all of it through fees. Each \$1 of deposits also saves the bank $s = \tau r_D = 0.20 \times 4\% = 0.008$ of taxes per year. Apart from these features, MM holds: the higher required return on equity at low $e$ only compensates shareholders for the higher risk of their claim, and the tax saving is paid by the government.

a. Find the annual failure probability $p(e)$ and the expected annual social cost of failure $F(e) = \kappa\, p(e)$ at $e = 4\%$ and at $e = 20\%$.

b. Explain why society's annual cost of the funding mix, per \$1 of assets and up to a constant, is

$$S(e) = \lambda e + \kappa \exp(-e/\theta),$$

and why neither the equity premium nor the tax saving $s$ appears in it. Derive the first-order condition, check the second-order condition, and find the socially optimal equity ratio $e^*$ in closed form and numerically. What is the annual failure probability at $e^*$? Compute $S(4\%)$, $S(e^*)$ and $S(20\%)$.

c. Per \$1 of assets per year, the bank's shareholders bear the private cost $(\lambda + s)\, e$ of using equity (fees and tax savings forgone on the deposits it replaces) and none of the failure cost. Which equity ratio does the bank choose if it is unregulated, and which if the regulatory minimum is $\underline{e} = 4\%$? For a move from 4% to $e^*$, compute the private cost to the shareholders and the social gain $S(4\%) - S(e^*)$.

d. A planner who believes two of the slide-47 statements computes a different "optimal" ratio. (i) Counting the lost tax saving as a social cost, she uses $\lambda + s$ in place of $\lambda$. (ii) Believing that equity is costly because of its higher required return, she uses the gap of 10 percentage points between a 14% required return on equity and a 4% deposit rate, 0.10 per \$1 of equity per year, in place of $\lambda$. Find the ratio each calculation gives and compare it with $e^*$.

e. Derive $\partial e^*/\partial\kappa$, $\partial e^*/\partial\lambda$ and $\partial e^*/\partial\theta$, and sign them at the given parameter values. By how much does $e^*$ rise if the social loss of a failure $\kappa$ doubles?

**Context needed to solve it** *From Prof. Ricca's Topic 2B slide 47 (Aula/Topic2B.pdf):*

[...] negative externality because the increase in leverage raises the probability of a bank failure, weakening the financial system and imposing losses on the broader economy.

*From Prof. Ricca's Topic 2B slides 44–45 (Aula/Topic2B.pdf) — the static trade-off this exercise transposes from the firm to society:*

An increase in $K$ has two effects. It increases the value of the tax shield [...] But it increases the expected bankruptcy cost $C$: $C\,H(K)$. The optimal debt level $K^*$ is such that the marginal benefit of debt equals the marginal expected bankruptcy costs. [...] With bankruptcy costs, $V_L = V_U + V_{ITS} - PV(\text{bankruptcy costs})$.

*Results of MM without and with taxes (Topic 2A slides 98–102; Topic 2B slides 7–8), restated:*

Without taxes, a change in leverage leaves $r_{WACC} = r_A$ and the value of the assets unchanged: the higher required return on equity at higher leverage, $r_E = r_A + \frac{D}{E}(r_A - r_D)$, is compensation for risk, not a cost to anyone. With corporate taxes, $V_L = V_U + \tau D$ for constant perpetual debt; the gain $\tau D$ is paid by the government, whose tax claim falls by the same amount, so the total value of the claims on the assets (shareholders, debtholders and government) does not depend on $D$.

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*Original exercise (list, Exercise 18), built on Topic 2B slides 44–47. All amounts per \$1 of bank assets per year.*

**What is asked.** (a) How likely failure is, and how much it costs society in expectation, at a thin and a thick capital ratio. (b) Society's cost of the funding mix, why only two terms belong in it, and the ratio that minimises it. (c) What the bank chooses on its own, and how its private cost of more capital compares with society's gain. (d) How the slide-47 fallacies (Statements 2 and 3) bias a planner's "optimal" ratio downward. (e) How $e^*$ responds to the social loss of a failure, the value of liquidity and asset risk.

**Approach.** This is the *static trade-off* of slides 44–45 seen from society's side: the marginal social benefit of debt (liquidity services) is set equal to the marginal expected cost of failure. By Exercises 16–17 (MM without and with taxes), neither the equity premium nor the tax shield is a social cost, so they are left out. The failure cost is a *negative externality* (slide 47): the bank does not bear it, which is why the private and social optima differ.

**Notation.**

- $e \in [0,1]$: equity ratio; $1 - e$: deposits per \$1 of assets; $\ell \geq 0$: annual loss rate, $\Pr(\ell > x) = \exp(-x/\theta)$, $\theta = 0.01$.
- $p(e) = \Pr(\ell > e)$: annual failure probability; $\kappa = 0.50$: social loss of a failure per \$1 of assets; $F(e) = \kappa p(e)$.
- $\lambda = 0.005$: liquidity value per \$1 of deposits per year; $s = 0.008$: tax saving per \$1 of deposits per year.
- $S(e)$: social cost of the funding mix; $e^*$: its minimiser; $\underline{e}$: regulatory minimum.

**Step 1 — (a) Failure probability and expected failure cost.** The bank fails when $\ell > e$, so

$p(e) = \exp(-e/\theta)$:

$$\begin{aligned} e = 4\%: \quad & p = \exp(-0.04/0.01) = \exp(-4) = 0.018316, \\ & F = (0.50)(0.018316) = 0.0091578, \\ e = 20\%: \quad & p = \exp(-0.20/0.01) = \exp(-20) = 2.06 \times 10^{-9}, \\ & F = (0.50)(2.06 \times 10^{-9}) = 1.03 \times 10^{-9}. \end{aligned} \tag{18.1}$$

**Answer (a).** $p(4\%) = 1.83\%$ a year and $F(4\%) = 0.916\%$ of assets a year; at 20% failure is practically impossible ($p = 2.06 \times 10^{-9}$, $F = 1.03 \times 10^{-9}$).

**Step 2 — (b) The social cost function.** Deposits yield a social benefit $\lambda(1-e) = \lambda - \lambda e$ a year, so relative to $e = 0$ each unit of equity forgoes $\lambda$ of liquidity services. Adding the expected failure cost and dropping the constant $-\lambda$:

$$S(e) = \lambda e + \kappa \exp(-e/\theta). \tag{18.2}$$

The equity premium is absent because, by MM (Exercise 16), it only compensates shareholders for the risk they bear and the total required return on the assets does not change. The tax saving $s$ is absent because it is a transfer from the government (Exercise 17): what the bank gains, taxpayers lose.

**Step 3 — (b) First- and second-order conditions.** Differentiate (18.2), using $\frac{d}{de}\exp(-e/\theta) = -\frac{1}{\theta}\exp(-e/\theta)$:

$$\begin{aligned} S'(e) &= \lambda - \frac{\kappa}{\theta}\exp(-e/\theta) = 0, \\ S''(e) &= \frac{\kappa}{\theta^2}\exp(-e/\theta) > 0, \end{aligned} \tag{18.3}$$

so the stationary point is the global minimum. Solve (18.3):

$$\begin{aligned} \exp(-e^*/\theta) &= \frac{\lambda\theta}{\kappa} \\ -\frac{e^*}{\theta} &= \ln\frac{\lambda\theta}{\kappa} \\ e^* &= \theta \ln\frac{\kappa}{\lambda\theta}. \end{aligned} \tag{18.4}$$

$e^* > 0$ because $\kappa/(\lambda\theta) > 1$. Numerically,

$$\frac{\kappa}{\lambda\theta} = \frac{0.50}{(0.005)(0.01)} = \frac{0.50}{0.00005} = 10{,}000, \qquad e^* = (0.01)\ln(10{,}000) = (0.01)(9.2103) = 0.092103.$$

By (18.4) the failure probability at the optimum is $p(e^*) = \lambda\theta/\kappa = 0.00005/0.50 = 0.0001$, once in 10,000 years.

**Step 4 — (b) Social cost at three ratios.**

$$\begin{aligned} S(4\%) &= (0.005)(0.04) + 0.0091578 = 0.0002 + 0.0091578 = 0.0093578, \\ S(e^*) &= (0.005)(0.092103) + (0.50)(0.0001) = 0.00046052 + 0.00005 = 0.00051052, \\ S(20\%) &= (0.005)(0.20) + 1.03 \times 10^{-9} = 0.0010000. \end{aligned}$$

**Answer (b).** $e^* = \theta\ln(\kappa/(\lambda\theta)) = 9.21\%$, with failure probability 0.01% a year; $S(4\%) = 0.93578\%$, $S(e^*) = 0.05105\%$, $S(20\%) = 0.10000\%$ of assets a year.

**Step 5 — (c) The bank's choice.** The shareholders' private cost $P(e) = (\lambda + s)e = 0.013\,e$ has $P'(e) = 0.013 > 0$ at every $e$: each extra unit of equity only costs them, and the failure cost it would

avoid falls on others. The bank picks the lowest ratio allowed: $e = 0$ unregulated, $e = \underline{e} = 4\%$ under the minimum. Moving from 4% to $e^*$:

$$\text{private cost} = (0.013)(0.092103 - 0.04) = (0.013)(0.052103) = 0.00067734,$$
$$\text{social gain} = S(4\%) - S(e^*) = 0.0093578 - 0.00051052 = 0.0088473.$$

Of the private cost, $(0.005)(0.052103) = 0.00026052$ is the liquidity loss (already inside $S$) and $(0.008)(0.052103) = 0.00041683$ is tax paid to the government.

**Answer (c).** The bank chooses $e = 0$ unregulated and exactly 4% under the minimum. The move to $e^*$ costs shareholders 0.0677% of assets a year and saves society 0.8847%, about 13 times more.

**Step 6 — (d) The planner's mistakes.** Each mistake replaces $\lambda$ in (18.4):

$$\text{(i)}\quad \lambda + s = 0.013: \quad e = (0.01)\ln\frac{0.50}{(0.013)(0.01)} = (0.01)\ln(3{,}846.15) = (0.01)(8.2548) = 0.082548,$$
$$\text{(ii)}\quad 0.10: \quad e = (0.01)\ln\frac{0.50}{(0.10)(0.01)} = (0.01)\ln(500) = (0.01)(6.2146) = 0.062146.$$

The failure probabilities rise to $(0.013)(0.01)/0.50 = 0.00026$ and $(0.10)(0.01)/0.50 = 0.002$, and the true social costs to $S = (0.005)(0.082548) + (0.50)(0.00026) = 0.00041274 + 0.00013 = 0.00054274$ and $S = (0.005)(0.062146) + (0.50)(0.002) = 0.00031073 + 0.001 = 0.00131073$, both above $S(e^*) = 0.00051052$.

**Answer (d).** (i) 8.25%, (ii) 6.21%, both below $e^* = 9.21\%$; the larger the cost wrongly assigned to equity, the thinner the "optimal" capital and the higher the true social cost.

**Step 7 — (e) Comparative statics.** Differentiate (18.4), written as $e^* = \theta\big[\ln\kappa - \ln\lambda - \ln\theta\big]$:

$$\frac{\partial e^*}{\partial \kappa} = \frac{\theta}{\kappa} = \frac{0.01}{0.50} = 0.02 > 0,$$
$$\frac{\partial e^*}{\partial \lambda} = -\frac{\theta}{\lambda} = -\frac{0.01}{0.005} = -2 < 0,$$
$$\frac{\partial e^*}{\partial \theta} = \big[\ln\kappa - \ln\lambda - \ln\theta\big] + \theta\Big(-\frac{1}{\theta}\Big) = \ln\frac{\kappa}{\lambda\theta} - 1 = 9.2103 - 1 = 8.2103 > 0.$$

Doubling $\kappa$ changes $e^*$ by

$$\theta\ln\frac{2\kappa}{\lambda\theta} - \theta\ln\frac{\kappa}{\lambda\theta} = \theta\ln 2 = (0.01)(0.6931) = 0.0069315,$$

from 9.21% to $(0.01)\ln(20{,}000) = 9.90\%$.

**Answer (e).** $e^*$ rises with $\kappa$ (+0.02 per unit) and with $\theta$ (+8.21 per unit, since $\kappa/(\lambda\theta) > e$), and falls with $\lambda$ ($-2$ per unit). Doubling $\kappa$ adds 0.69 percentage points.

**Intuition.** The bank's leverage choice is a trade-off in which it bears all of the benefit of debt and none of the cost of failure. Society's trade-off counts both, and once the MM "costs" of equity are seen to be risk compensation and transfers, the only real cost of equity left is the small loss of liquidity services. That is why the social optimum is well above what banks would choose, and why getting Statements 2 and 3 wrong biases capital requirements downward.

**Common mistakes.**

- Putting the equity premium or the tax shield into the social cost; both drop out by MM.
- Stating the first-order condition without differentiating $\exp(-e/\theta)$ (the factor $-1/\theta$ is where $\theta$ enters $e^*$).
- Reading the small private cost in (c) as proof that capital is cheap for society, or the reverse: the private cost includes a transfer, the social gain is the externality avoided.

**Ref:** Prof. Ricca, Topic 2B slides 44–45 (static trade-off: marginal benefit of debt = marginal expected bankruptcy cost, $V_L = V_U + V_{ITS} - PV$(bankruptcy costs)), slides 46–47 (bank capital debate, failure externality).